What Does Reinforcement Learning Add to a Small Translator for African Languages?

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Abstract

Small language models translate many African languages poorly, and one common failure is that they do not finish their sentences: they stop midway or repeat themselves. We train a small English-to-African translation model with supervised fine-tuning, then with reinforcement learning that rewards outputs rated well by automatic quality scorers, and we ask what the second step adds. It improves every language we test, by about nine points of the standard chrF++ score on one benchmark. About half of that gain is finishing sentences and not repeating; a single decoding setting gives the fine-tuned model the same half without any extra training, while the rest of the gain remains. Spending the same computing time on more fine-tuning helps only the languages for which we supplied extra data, and by the rule we fixed beforehand the comparison is inconclusive. For scientific text, the extra fine-tuning step, not the reinforcement learning, carries the gain. On identical test sentences, our model stays about one point below NLLB-600M and about three below NLLB-3.3B (about 92% of its score), which is about 8.6 times larger, and a model distilled from NLLB-3.3B is ahead of ours on its own languages. We trained one model per setting, checked seed noise on a subset only, and used automatic scores only; no person judged the outputs.

1 Introduction

Machine translation into African languages is hard for small models. Training text is scarce, and small models have little capacity to spare. Large systems such as NLLB-200 (NLLB Team et al., 2022) help, but they are big, and the NLLB models and their outputs carry non-commercial licence terms.

Small models also fail in a characteristic way. After supervised fine-tuning (SFT), our model often does not finish its sentences: it stops midway, repeats a phrase until it runs out of space, writes in the wrong language, or copies the English input. Reinforcement learning (RL) with automatic quality scores as rewards is a natural remedy, and recent work reports gains for translation (Shao et al., 2024; Feng et al., 2025; Finkelstein et al., 2026). Those gains, however, may come from something simpler. Repeating-and-not-stopping is partly a property of greedy decoding, which a decoding setting can change; and RL costs computing time that more SFT could use instead.

We therefore ask four questions about a 385-million-parameter English-to-African translator, trained first with SFT and then with RL:

- **RQ1.** How much does RL improve translation, and for which languages?
- **RQ2.** How much of the gain is finishing and not repeating, rather than better translation, and can a decoding change provide it?
- **RQ3.** Does the same computing time spent on more SFT do as well?
- **RQ4.** How reliable is the gain, and how does the model compare with a much larger system?

Findings at a glance.

- RL improves every language we test on three standard benchmarks (RQ1).
- About half of the gain is finishing sentences and not repeating. A decoding setting recovers that half for the SFT model without training, but RL still adds clear gains beyond it (RQ2).
- Decoding tricks such as beam search and minimum Bayes risk (MBR) decoding help the SFT model a lot and the RL model little (RQ2).
- Matching the compute with more SFT is inconclusive by our pre-registered rule: it matches RL on the 19 languages where it had extra data, and is far behind on the others (RQ3).
- Almost all of the gain arrives in the first RL stage (RQ4). On identical sentences the model is still about one chrF++ point below NLLB-600M and about three below NLLB-3.3B, about 92% of its

score at about one ninth of its size; a model distilled from NLLB-3.3B is ahead of ours on its languages (Section 5.1).

Contributions. (1) A controlled decomposition of an RL gain in small-model translation: a decoding control, a matched-compute control, seed and ablation runs, and an independent audit of the reward's own repetition check. (2) A decontaminated SFT and RL pipeline for 125 languages, with an audit of the training text. (3) Negative results: RL aimed at weak languages, science fine-tuning on a quality-estimation metric, and whole-document translation. Model artefacts: model weights at `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl` (see Ethical considerations). We do *not* claim that RL beats more fine-tuning in general, that our model beats NLLB, or any human-rated quality: all scoring is automatic.

2 Related Work

Translation for African languages. Community benchmarks and corpora made progress measurable: Masakhane (Nekoto et al., 2020), MAFAND-MT (Adelani et al., 2022), FLORES-200 and its community successor FLORES+ (Goyal et al., 2022; Open Language Data Initiative, 2025), NTREX-128 (Federmann et al., 2022), SALT (Akera et al., 2022) and SMOL (Caswell et al., 2025). NLLB-200 (NLLB Team et al., 2022) is the reference system. Recent language models for African languages are much larger than ours (Yu et al., 2026; Akera et al., 2025); TranslatePsy-AfriSLM (Gritta et al., 2026) is the nearest small competitor. For science text, AfriScience-MT (Abdulmumin et al., 2026) uses SSA-COMET (Li et al., 2025) as its main metric.

Reinforcement learning for translation. GRPO (Shao et al., 2024; Yu et al., 2025) powers recent translation systems with reasoning-style training (Feng et al., 2025; Wang et al., 2025; Cheng et al., 2025; Zheng et al., 2025), and TranslateGemma rewards with an ensemble of quality models (Finkelstein et al., 2026). Closest to us, other work trains NLLB models with round-trip or reference-free rewards (Attia and Aji, 2026; Garcia-Estrada et al., 2026) and studies reward hacks in quality-estimation rewards (Liu et al., 2026; Han et al., 2026). Whether RL generalises beyond what supervised training memorises has been asked for reasoning (Chu et al., 2025; Yue et al., 2025). For translation, we searched for a comparison at matched computing time and found none; the nearest compares reference-free RL with three epochs of supervised training (Garcia-Estrada et al., 2026).

Decoding and evaluation. Minimum Bayes risk and quality-aware decoding (Freitag et al., 2022; Fernandes et al., 2022) aim at similar gains without training, so we measure them as controls. AfriCOMET (Wang et al., 2024), chrF++ and sacreBLEU (Popović, 2017; Post, 2018) and the paired bootstrap (Koehn, 2004) are our metrics and tests. Learned rewards are easy to over-optimise (Gao et al., 2023; Coste et al., 2024), which is why we keep the metrics we report separate from the metrics we reward. We claim no priority for RL in translation, for small models, or for language coverage.

3 Approach

Figure 1 shows the pipeline. We describe the model and the two training steps, then the variants used as controls and extensions.

Model. We start from Gemma-3-270m-it (Gemma Team et al., 2025) and add layers to reach 385M parameters (39 layers) (cf. Wu et al., 2024; Kim et al., 2023). The task is English to one of 125 languages. A single extra token in the vocabulary names the target language, and the model answers with that token followed by the translation.

Step 1: supervised fine-tuning (the SFT baseline). We train on English-to-African sentence pairs and instruction data. Before training, we remove every pair that matches the text of any benchmark we later use, checked on both sides of the pair. Part of the data comes from NLLB-mined sentences and a collection of translated instructions; we report their share and licences in the Ethical considerations. Training uses the Muon optimiser (Liu et al., 2025) and takes about 45 hours on one A100 GPU (Appendix C).

Step 2: reinforcement learning (the RL model). We use GRPO (Shao et al., 2024). For each English text the model samples 16 translations. Each translation receives a reward, and the model is nudged towards translations that score above the average of their group and away from those below it; a penalty keeps the model close to its starting point. Three stages follow one another, and the first does most of the work (Section 5.5). The reward has three parts. (i) A trans-

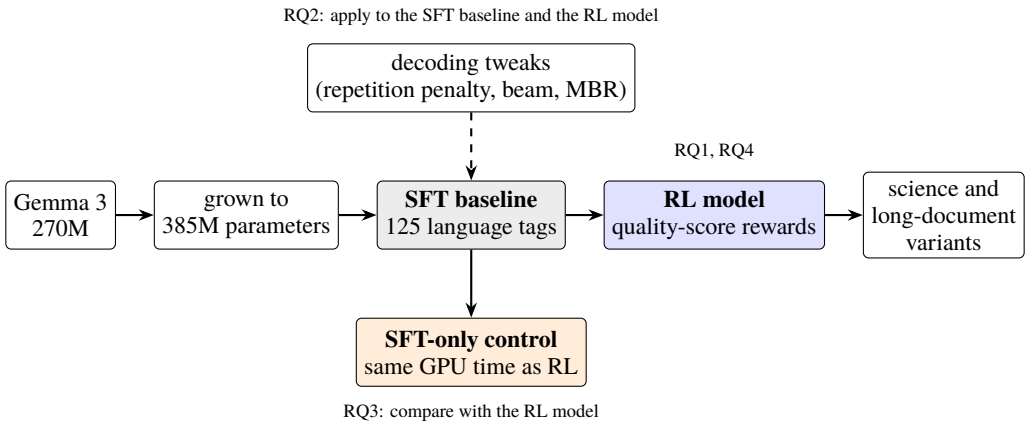


Figure 1: Overview. A 270M model is grown to 385M parameters and fine-tuned (the SFT baseline), then trained with reinforcement learning (the RL model). Two controls ask what else could give the same gain: decoding changes (RQ2) and more fine-tuning for the same GPU time (RQ3).

lation that fails a hard check scores the minimum: wrong or missing language tag, no end of text, a copy of the input, repetition, the wrong language, or an implausible length. (ii) Otherwise a quality score without a reference translation rates the output. We combine two quality models (Wang et al., 2024; Li et al., 2025), taking their mean unless they disagree strongly, and for languages without a validated quality model we reward agreement with the group's other samples. (iii) Small penalties discourage, among other things, unusual length, missing numbers or names, and missing diacritics. Exact settings are in Appendix C.

Variants. The *SFT-only control* continues SFT from the baseline for the same GPU time as the RL stages, using half replay data and half extra English-to-X pairs for 19 languages (Gritta et al., 2026). The *science variant* adds a short fine-tuning step on science text from AfriScience-MT (Abdulmumin et al., 2026) and then a second RL run; the *long-document variant* adds RL on long inputs. Internal names used in the appendix tables are listed in Appendix A.

4 Experimental Setup

Benchmarks. We translate English into African languages on the standard FLORES+ devtest set (Open Language Data Initiative, 2025) (1,012 sentences; 54 languages), NTREX-128 (Federmann et al., 2022) (24 languages) and MAFAND-MT (Adelani et al., 2022) (15 languages). Smaller sets, SmolSent (Caswell et al., 2025), SALT (Akera et al., 2022), GlotStoryBook (Kargaran et al., 2023), IbomMT (Kalejaiye et al., 2025) and FLORES-MOZ (MOZNLP, 2026), appear in the appendix, as does AfriScience-MT (Abdulmumin et al., 2026) for science text. Unless stated, we report the average over languages, with each language counting equally.

Metrics. *chrF++* compares character and word n-grams of the output with a human reference; higher is better. *Held-out AfriCOMET* is a learned score that compares source, output and reference; we never use it as a reward. The *finished rate* is the share of outputs that end properly; the *loop rate* is the share that repeat themselves. Both are the checks our reward also uses, so they are not independent evidence of better translation. We computed chrF++ with sacreBLEU 2.6.0 (word order 2, otherwise defaults) after Unicode normalisation; the full signature string was not saved.

Avoiding self-deception. First, *independence*: the metrics inside the reward (the two quality models, agreement between samples, and the finished, repetition and language checks) are not evidence that RL helped. Our evidence is chrF++ and held-out AfriCOMET against human references (Appendix M). Second, *paired tests*: every difference is a paired bootstrap over test sentences with a 95% interval, which covers sentence variance but not training randomness. Third, *contamination*: besides filtering the training pairs, we scanned 26 million training samples against all benchmark text and found no benchmark sentence of three or more words (Ethical considerations list the residuals). Fourth, *seeds*: every model is one training run; we measured seed noise only for the first RL stage on 20 languages.

Decoding. All main results use greedy decoding with up to 768 new tokens (vLLM, bf16). The *repetition penalty* used as a control is a decoding setting that discourages repeated tokens; we chose its strength on a development set by a rule fixed

in advance. Beam search and MBR decoding are described where they appear.

5 Results

5.1 How does the model compare with other systems?

Answer: the RL model beats the SFT baseline everywhere, but it stays below the NLLB systems. On identical FLORES+ sentences it scores about one chrF++ point below NLLB-600M and about three below NLLB-3.3B, which is about 92% of the larger system's score (Table 1, Figure 2). Held-out AfriCOMET gives the same order with a wider gap. The RL model is ahead of NLLB-600M in only some languages and of NLLB-3.3B in a few (last column). NTREX and MAFAND-MT show the same order against NLLB-3.3B; there, only language-level NLLB averages exist, so these rows have no sentence pairing and no AfriCOMET.

The 19 languages that TranslatePsy-0.8B covers show the limits of the comparison. TranslatePsy-0.8B is the strongest system there and clearly ahead of the RL model. It was distilled from NLLB-3.3B outputs, so its lead is partly a student–teacher effect and not independent evidence of a better design. The SFT-only control, which was trained on NLLB-3.3B translations for these languages, matches the RL model in chrF++ but not on the neural score. We decode greedily and NLLB with beam search (four beams); on a 20-language check, beam search or MBR moved the RL model by less than one chrF++ point either way (Section 5.3). SmolSent, SALT and three smaller benchmarks have no RL model; their per-language results are in Appendix B.

5.2 RQ1: How much does RL improve translation?

Answer: a lot, for every language we test. The RL model beats the SFT baseline on all three benchmarks (Table 2). Every language on FLORES+ and NTREX improves, and so does every MAFAND-MT language except Nigerian Pidgin, which does not change measurably. The gain is large where the baseline was weakest. On FLORES+ the baseline leaves about one output in six unfinished and repeats itself in about one in five; the RL model almost always finishes and rarely repeats (Table 2, lower rows).

Where RL does not help. For Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, quality stays very low (chrF++ below about 14) and RL makes one problem worse: the model increasingly answers in English. For Tunisian Arabic, English output rises from about one fifth of outputs to almost half (Appendix G). These languages were not among the RL prompts.

5.3 RQ2: Is the gain just finishing and not repeating?

Answer: about half of it is, and a decoding setting recovers that half for the SFT baseline; RL still adds clearly more. We tested this in three ways. First, we cut the baseline's output at the first repetition, which leaves the translation untouched; this recovers 49% of the chrF++ gain. Second, we gave the baseline a repetition penalty during decoding, with no further training. It recovers the same 49%. With the penalty on both models, RL is still about 4.4 chrF++ ahead (Table 3, Figure 3). This control is imperfect: the best penalty strength was at the edge of the range we searched, and stronger values looked better on a development set but were never evaluated fully, so the baseline may be undertuned.

Third, we tried stronger decoding. Beam search and MBR help the baseline much more than they help the RL model. MBR gives the baseline a large boost and the RL model a small one; for beam search, the RL result depends on how the search decides to stop, so we draw no conclusion from it. RL with plain greedy decoding stays about 1.6 points ahead of the best decoded baseline, which needs about 17 times the computation per sentence. The beam-search numbers of an earlier study that used 256 output tokens are not comparable, and we do not use them (Appendix I).

A caveat on our own checks. The repetition flag we report is also part of the reward, so we remeasured with a detector that shares none of its code. It finds about 1.8 times as many repetitive outputs as the reward's flag does. Rescoring the extra outputs at their language's average changes the RL model's chrF++ by at most 0.2, so the conclusion stands (Appendix G).

5.4 RQ3: Does the same compute on more SFT do as well?

Answer: inconclusive by the rule we wrote down beforehand. We continued SFT for the same GPU time as the three RL stages, on half replay data and half extra English-to-X pairs for 19 languages. The

system	chrF++	AfriCOMET	RL − system, chrF++	RL − system, AfriCOMET	RL ahead in
FLORES+, 44 languages, identical sentences (42,680)					
SFT baseline	25.6	0.550	+9.30 [+9.00, +9.55]	+0.073 [+0.071, +0.075]	44 of 44
SFT-only control[a]	28.6	0.586	+6.32 [+6.11, +6.52]	+0.038 [+0.036, +0.039]	34
RL model	**34.9**	**0.623**			
NLLB-600M	35.9	0.662	−1.05 [−1.24, −0.86]	−0.038 [−0.040, −0.036]	15
NLLB-3.3B	38.1	0.677	−3.21 [−3.38, −3.02]	−0.054 [−0.055, −0.052]	7
FLORES+, the 19 TranslatePsy languages (17,380 sentences)					
SFT baseline	34.0	0.589	+7.04 [+6.64, +7.41]	+0.054 [+0.051, +0.056]	
SFT-only control[a]	41.0	0.665	+0.08 [−0.13, +0.30]	−0.023 [−0.024, −0.021]	
RL model	**41.0**	**0.642**			
NLLB-600M	42.6	0.688	−1.54 [−1.78, −1.28]	−0.046 [−0.048, −0.043]	
NLLB-3.3B	44.7	0.701	−3.62 [−3.84, −3.39]	−0.059 [−0.061, −0.056]	
TranslatePsy-0.8B[b]	45.3	0.717	−4.23 [−4.39, −4.07]	−0.075 [−0.077, −0.073]	
NTREX (21 languages), MAFAND-MT (14); NLLB = language-level averages					
NTREX: SFT baseline	31.0	0.551	+7.97 [+7.73, +8.21]	+0.063 [+0.061, +0.065]	
NTREX: SFT-only control[a]	35.9	0.609	+3.14 [+2.97, +3.32]	+0.005 [+0.004, +0.006]	
NTREX: **RL model**	**39.0**	**0.614**			
NTREX: NLLB-3.3B	43.3	–	−4.26 [−5.22, −3.24][c]	–	1 of 21
MAFAND-MT: SFT baseline	34.5	0.614	+5.39 [+5.15, +5.63]	+0.040 [+0.038, +0.041]	
MAFAND-MT: SFT-only control[a]	39.5	0.670	+0.34 [+0.16, +0.52]	−0.016 [−0.017, −0.014]	
MAFAND-MT: **RL model**	**39.9**	**0.654**			
MAFAND-MT: NLLB-3.3B	44.6	–	−4.73 [−6.56, −3.12][c]	–	1 of 14

Table 1: Our models against other systems. chrF++ and held-out reference-based AfriCOMET: higher is better; averages over languages; 95% paired-bootstrap intervals for the RL model minus each system (positive = RL better), over test sentences. The 44 FLORES+ languages exclude the Arabic dialects, Tamazight, Dinka and Tamasheq, so averages are higher than elsewhere. [a]Trained on NLLB-3.3B translations for 19 languages (never released). [b]Distilled from NLLB-3.3B outputs. [c]Interval from a bootstrap over languages (no per-sentence NLLB output). Ours decode greedily; NLLB with 4 beams. Other benchmarks and per-language results: Appendix B.

Table 2: RL model against the SFT baseline ("baseline"). chrF++ (top rows, per benchmark with the number of languages) and AfriCOMET: higher is better. Averages over languages, greedy decoding, 95% paired-bootstrap intervals in brackets. One training run per model.

	baseline	RL model	gain
FLORES+ (54 lang.)	22.85	31.66	+8.81 [+8.59, +9.03]
NTREX (24)	29.25	37.17	+7.91 [+7.69, +8.12]
MAFAND-MT (15)	35.77	40.78	+5.01 [+4.76, +5.23]
all three, all lang.	26.59	34.55	+7.96 [+7.82, +8.12]
AfriCOMET, all three	0.565	0.628	+0.063 [+0.062, +0.064]
FLORES+: ends properly	83%	99.5%	
FLORES+: repeats	21%	5%	

Table 3: chrF++ with and without decoding changes (higher is better; 95% intervals for the RL gain). Top: all three benchmarks. Bottom: a 20-language subset with 200 sentences each. Repetition penalty = a decoding setting that discourages repeated tokens; MBR = choose the sample most similar to the others.

decoding	SFT baseline	RL model	RL gain
greedy	26.6	34.6	+7.96 [+7.82, +8.12]
+ repetition penalty	30.5	34.8	+4.37 [+4.28, +4.46]
greedy (20 lang.)	32.3	39.9	+7.56 [+6.89, +8.19]
beam, 4 beams	36.0	39.1	+3.07 [+2.67, +3.46]
MBR, 16 samples	38.3	40.8	+2.52 [+2.26, +2.80]

Table 4: SFT-only control against the RL model, chrF++ (higher is better), averaged over language–benchmark pairs, 95% paired-bootstrap intervals. The control received extra data for 19 languages.

languages	control	RL model	RL − control
all	29.97	34.55	+4.58 [+4.45, +4.70]
19 with extra data	41.37	41.34	−0.03 [−0.13, +0.09]
all others	18.32	27.61	+9.29 [+9.09, +9.49]

extra pairs are NLLB-3.3B translations of web-text sentences (Gritta et al., 2026). We fixed the decision rule, the main comparison and the language groups before looking at any benchmark result for the control. The rule had four conditions; three were met. The fourth, that RL must also beat the control on the 19 languages that had extra data, was not met. So by our rule the result is inconclusive, and we did not change the rule afterwards.

In plain terms (Table 4, Figure 4): on the 19 languages with extra data, the control matches RL in chrF++ and finishes its sentences almost as reliably; it is even ahead on the learned metrics there, which were partly used to filter its data. On the other languages, the control barely improves on the baseline, while RL improves them all. Taking only the overall number, RL looks better; taking only the 19 languages, the two look equal. Neither alone is the registered verdict.

Caveats. The control is one run with one learning rate and one stopping point; a different mix or a longer run could change the picture. It is also not data-matched: its extra pairs come from a larger system (NLLB-3.3B). We therefore claim neither that RL is a better use of compute than SFT nor

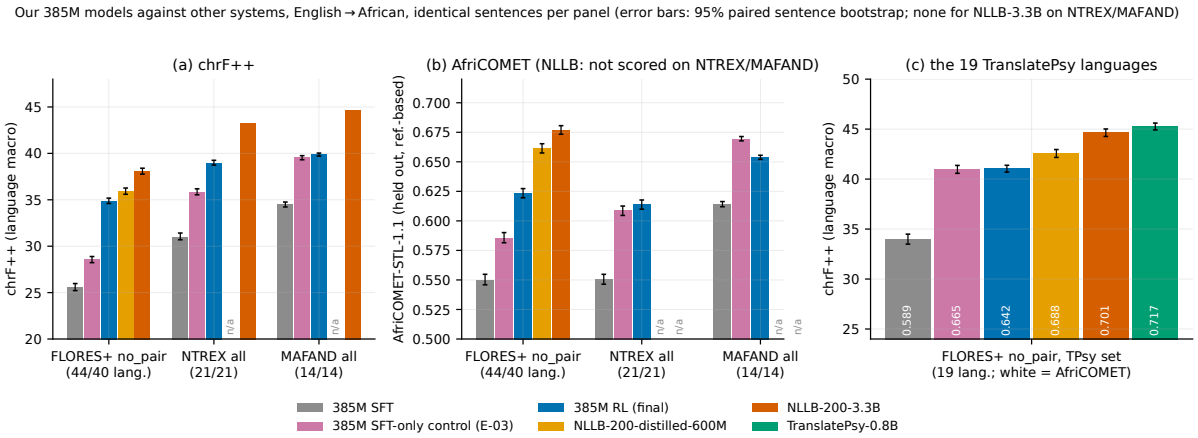


Figure 2: Our models against other systems (language averages on identical sentences; higher is better). (a) chrF++ and (b) held-out AfriCOMET on FLORES+, NTREX and MAFAND-MT; NLLB has no AfriCOMET score and no per-sentence output on NTREX and MAFAND-MT. (c) The 19 TranslatePsy languages, with AfriCOMET printed in the bars. **The vertical axes do not start at zero.** Error bars: 95% paired bootstrap over sentences; none for NLLB-3.3B on NTREX and MAFAND-MT. "SFT-only control" was trained on NLLB-3.3B translations for 19 languages.

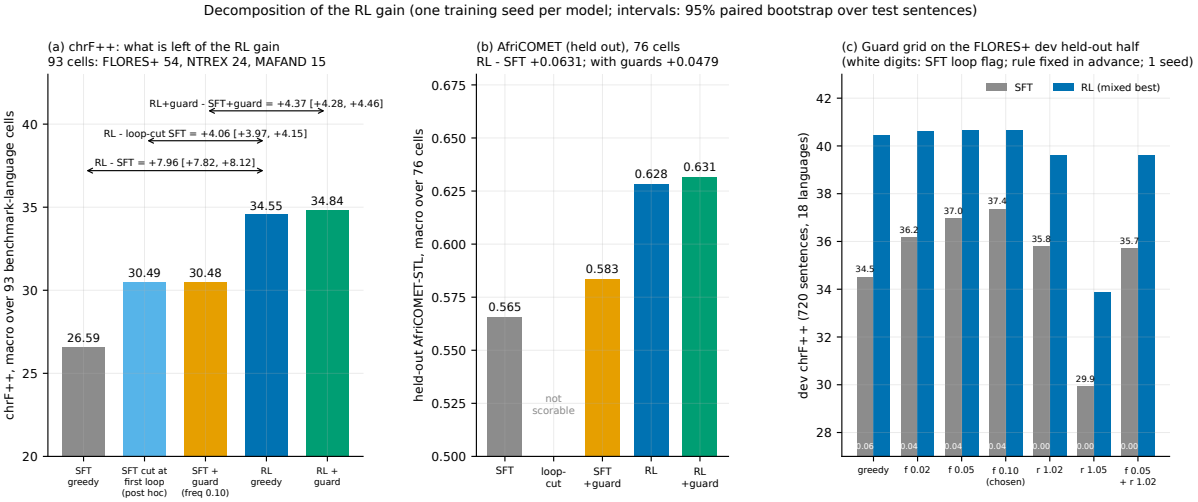


Figure 3: What is left of the RL gain once finishing is fixed. (a) chrF++, higher is better, averaged over all three benchmarks: the SFT baseline, the baseline with its repeated tails cut off, the baseline with a repetition penalty, the RL model, and the RL model with the same penalty. Arrows show paired differences with 95% intervals. (b) The same for held-out AfriCOMET. (c) How the penalty strength was chosen on a development set; the chosen value is at the edge of the grid.

the opposite.

5.5 RQ4: How reliable is the gain, and how does the model compare with a larger system?

Answer: the gain is robust to random seeds, nearly all of it comes from the first RL stage, and the model reaches about 92% of NLLB-3.3B's chrF++ on shared languages (Section 5.1). The first RL stage produces 94% of the gain reached after the second stage, and the two later stages add little. Two extra random seeds of the first stage, run on 20 languages, differ by only 0.06 chrF++, far less than the gain itself. The last RL step is about the size of that seed noise, so we do not call it a gain (Figure 5).

We also tested four features that we added during the first stage. Together they are worth about 0.3 chrF++ at 100 steps, mostly because of one of them, a term that rewards agreement with human reference translations for seven languages. A beam-search guide, which costs about a quarter more time per step, is not shown to be worth it, and none of the features causes the drift from the starting model.

NLLB-3.3B is about 8.6 times larger than our model, and Section 5.1 gives the comparison. The RL model closes about three quarters of the gap that the baseline leaves to NLLB-3.3B, and the remaining gap is not a matter of unfinished sentences. On the 43 languages shared with the published NLLB table the RL model scores about 92%

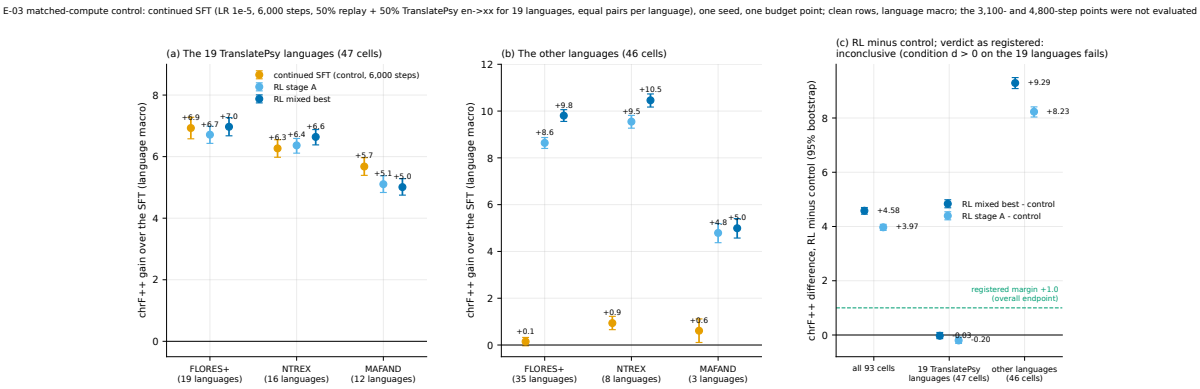


Figure 4: Gain over the SFT baseline in chrF++ (higher is better) for the SFT-only control, the first RL stage and the final RL model, on the 19 languages with extra control data (a) and on the others (b); (c) RL minus control, with the +1.0 margin of the registered rule. Error bars: 95% intervals. One run each; only the final control checkpoint was evaluated.

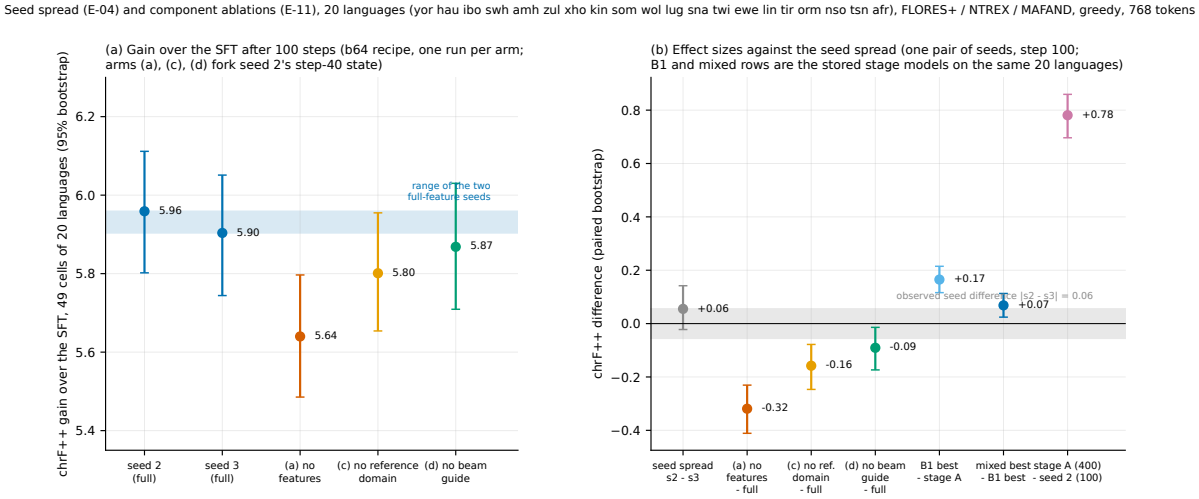


Figure 5: Seed noise and what four extra training features are worth (20 languages, chrF++ gain over the SFT baseline after 100 RL steps). (a) Two seeds of the full recipe (shaded band) and three runs that each remove a feature. (b) The same effects against the seed difference (shaded): removing all four features costs about 0.3 points, removing the beam-search guide is not separable from seed noise. Error bars: 95% intervals; one run per arm.

of NLLB-3.3B's chrF++ (Appendix B). It matches or beats NLLB-3.3B on seven languages, all among those where NLLB scores lowest. Caveats: NLLB's published numbers use a slightly different test set and beam search; the hardest languages have no published NLLB number; and for a few languages the reward itself used NLLB outputs as a target. We therefore do not claim to beat NLLB. Compared with our earlier 270M model run in the same harness, the 385M SFT baseline is behind, the RL model is level on two benchmarks and behind on one, and it is behind at document level (Appendix B).

5.6 Further results: science text and long documents

Answer: for science text, the extra fine-tuning step carries the gain, and whole-document translation still does not work. A short fine-tuning step on science text raises chrF on AfriScience-MT but not the neural score the benchmark uses. An RL run started from that model adds a little more. To ask whether the science step is needed, we ran the same RL without it (Table 5): it ends about 1.4 chrF below the model that had the science step (one run each), and it gains little over its own start. The neural quality score rates the two models as equal, although chrF and held-out AfriCOMET rate the no-science-step model lower; because that score belongs to the family of scores the RL optimises, we do not treat it as evidence of quality. Translating whole documents in one pass fails for all our models. RL on long inputs raises the share of finished documents from 70% to 82%, but about three quarters of the documents still contain loops, so we translate sentence by sentence (Appendix K).

Table 5: Science text (AfriScience-MT test sentences, six languages, one run each). Higher is better. The last row is 1.44 chrF $[1.25,\ 1.61]$ below the third row.

model	chrF	AfriCOMET
RL model	45.4	0.683
+ science fine-tuning	46.6	0.688
+ science fine-tuning, then RL	47.1	0.693
RL without science fine-tuning	45.7	0.685

6 Discussion and Conclusion

What the evidence says. RL improves a small translator on every benchmark we use, but the gain has two parts that deserve separate names. About half is finishing sentences and not repeating, which a decoding setting also gives the fine-tuned model; the other half is better translation that the setting does not provide. Almost all of it arrives in the first RL stage. More fine-tuning for the same time is a serious alternative where it has extra in-language data, and a much weaker one where it does not; our registered comparison is therefore inconclusive, and what RL adds in our data is breadth, since it improved every language from the model's own samples and a quality score. For science text, the fine-tuning step, not RL, carried the gain.

What we do not claim. We do not claim that RL is a better use of compute than SFT in general, that the improvement beyond finishing is quality a human would confirm, that our model beats NLLB or our earlier 270M model, or that any reward ingredient other than the four tested features helped. The model is not usable for Arabic dialects, Tamazight, Dinka, Tamasheq or Kanuri.

What would settle the open points. Evaluating the stronger repetition penalties and the two unevaluated control checkpoints, more seeds for the later stages, and human judgement of outputs. Together with the licence work in the Ethical considerations, these are the next steps.

Limitations

Automatic scoring only. The references of FLORES+, NTREX-128 and MAFAND-MT are human translations, but all scoring is automatic: no human judgement of system outputs was collected. The neural scores belong to one family, and AfriCOMET has a length and entity bias. **One training run per model.** We measured seed noise only for the first RL stage, on 20 languages and 100 steps, with one pair of extra seeds. The last RL step and per-language differences below about half a point cannot be separated from seed noise. A second seed of the SFT baseline is impractical (about 45 GPU-hours). **The matched-compute control is one run:** one seed, one learning rate, one data mix with NLLB-3.3B targets and one evaluated stopping point. The repetition-penalty control chose its best value at the edge of the range searched. **Rewarded metrics are not evidence.** The finished, repetition and language checks, and the quality scores the RL optimises, are part of the reward. The independent repetition detector shares the idea of the reward's check and cannot see non-repetitive failures. **Ablations are limited:** three arms, 100 steps, one run each, 20 languages; later RL stages and most reward ingredients were not ablated, and mid-run changes (Ethical considerations) mean the stages are not one clean design. **Decoding and engine.** Headline numbers use one inference engine in bf16. The two engines differ in bf16 but agree in fp32; we did not run the RL model in fp32. Beam search and MBR cover 20 languages and one beam width. **Coverage and comparisons.** Fifty-one of the 125 languages have no reference test and are unscored. The model fails on Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, where RL increases English output. Comparisons with the earlier 270M model and with NLLB mix differences in data, steps, test sets and decoding: our decoding is greedy, NLLB's is beam search, and on NTREX and MAFAND-MT only language-level NLLB averages exist (no sentence pairing, no AfriCOMET). Document results rest on 28 papers per language, and whole-document translation remains unusable. **Not run.** Human judgement of outputs; the NLLB-data ablation; scans of packed training shards for contamination; a 54-language decoding study; a science fine-tuning step at full strength followed by RL; a second seed of the science run; long-document RL from the no-science-step model.

Ethical considerations

Data and licences. Part of the fine-tuning data is derived from NLLB. Sentences mined with NLLB make up about 14% of the tokens of the main fine-tuning stage; adding NLLB-Seed and the Aya translations (NLLB-3.3B outputs) gives about 19%, and adding machine-translated Tumbuka instructions (system not stated, possibly NLLB) about 23%. NLLB models and outputs carry non-commercial (CC-BY-NC) terms, which limits commercial use of the NLLB-derived data and signals; a licence statement must accompany any release of the weights. Several sources have unspecified or unconfirmed licences (the `*_mt560` OPUS sets, which are mostly Bible and Jehovah's Witnesses text and 69% of the main pair data; `multi-open`; AfroLingu-MT; an Efik New Testament sheet) and need a legal decision. IbomMT is CC-BY-NC-4.0 and is a disclosed exception. Religious text dominates many low-resource languages. Reward-side data: NLLB-3.3B outputs (CC-BY-NC) served as a reward target for five languages only, and RL prompts (FineWeb-Edu, Wikipedia, news including BBC News, which declares no licence and carries BBC copyright) are input text only. The Gemma terms of use apply to the derived weights, and pre-training exposure cannot be excluded. Reward and monitor models, as declared on the Hub (read 2026-11-05): the two SSA-COMET and the two AfriCOMET checkpoints are apache-2.0; GlotLID declares `license: other` with `license_name: apache-2.0-plus-notices`.

Release and licence. The final RL model (branch `main`) is at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl, the SFT model at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft, and the seed, ablation and science-variant weights at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments; the control checkpoint is not released. At the last check the Hub repositories declare no licence field and have no model card; the Gemma terms of use apply, and the NLLB-derived and non-commercial data above limit commercial use **[OWNER: confirm release licence]**. The data repositories named in our reports were private at the last check **[OWNER: which parts of the data are released]**.

The control's extra data. For the SFT-only control we used TranslatePsy English-to-X pairs (CC-BY-NC-4.0; NLLB-3.3B translations of web sentences, filtered by AfriCOMET, SSA-COMET and MetricX) for that experiment only. No TranslatePsy row is in the SFT baseline, any RL stage, the science or long-document variants, or any released weight; the control checkpoint is neither released nor used in production. A good control result partly means "distilling NLLB". TranslatePsy-AfriSLM-0.8B and -2B appear only as comparison systems; they were distilled from NLLB-3.3B outputs. The NLLB-218 language-identification model (CC-BY-NC) was used for measurement only.

Decontamination and its residuals (the clean 385M pipeline). The audit found no benchmark sentence of three or more words in the audited training text. The remaining benchmark matches are one- or two-word strings, numerals and names (for example 0.25% of NTREX rows and 0.47% of MAFAND-MT rows). The FLORES+ development half (440 sentence ids) is in training on purpose, and the held-out half is protected. A lexical fuzzy scan of the pair and instruction files found no benchmark string of eight or more tokens as a contiguous run, but strings of four to seven tokens occur inside longer training text, and 263 benchmark strings of eight or more tokens overlap training text by character 5-grams; these are candidates, and we did not measure their effect. The first RL stage drew 6 flagged prompts of 28,120 from seven unfiltered prompt files, and 0.9% of AfriScience-MT training rows have their African-language sentence in the earlier pair data (the test papers are untouched). We did not run paraphrase, machine-translation round-trip or packed-shard scans. These statements are about the clean 385M pipeline. Our earlier 270M pipeline trained on text that held the English side of 95% of FLORES+ devtest; its numbers appear only on rows clean for both pipelines.

Analysis decisions and process. We chose checkpoints on a 720-sentence FLORES+ development subset, chose the languages for the weak-language run from devtest-derived tables, ran the decomposition and family analyses after seeing results, and changed the science headline metric after writing the interpolation rule. The authors overrode two failed training gates. The first RL stage was stopped after 40 steps and resumed with four added features; later stages were restarted once or twice, and the hacking alarm was loosened once. The science RL step cap was extended by the trainer, one long-document run lost most of its time to a disk-full crash loop, and the no-science-step run had one out-of-memory restart (replayed steps use the later records). LibreTexts rows were removed from the long-document pool before it started.

Intended use, risks and dual use. Outputs are machine translations of unverified quality, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable, RL increases English output there, and documents should be translated sentence by sentence. The model should not replace NLLB-class systems or be used where mistranslation is costly (health, legal, safety). As with any translation model, it can produce text at scale in under-served languages, including spam or misleading content, and we release no content filter.

Author contributions

[Placeholder, to be filled by the lead author: one sentence per person describing what that person contributed (for example design, data, training, evaluation, writing). Authorship is based on contribution.]

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A Glossary of internal names

The main text uses descriptive names. The appendix tables use the internal names of our experiment log; this table maps them.

internal name	name in the main text	meaning
A0	earlier 270M model (base of the grown model)	the 270M model with six inserted layers, trained before growth
SFT-1, SFT-2, stage B	SFT baseline (training steps)	SFT-1 trains only the new layers and language tags; SFT-2 (stage B) trains all weights
SFT, SFT final	SFT baseline	the 385M model after supervised fine-tuning
stage A	first RL stage	400 steps on short English windows
B1 best, B1 final	second RL stage (best checkpoint at step 50; final at step 135)	RL on longer windows; best and final differ by less than the noise (E-20)
mixed, mixed best	third RL stage; **RL model**	short and long windows; its best checkpoint is the final general RL model used throughout
weak-group RL	RL aimed at eight weak languages	85 extra steps; no measurable gain on its targets
science SFT, α	science fine-tuning step; + *science fine-tuning*	fine-tuning on science text, then weight interpolation between the starting model ($\alpha = 0$) and the fine-tuned model ($\alpha = 1$); $\alpha = 0.5$ is used
science RL	science RL	RL started from the science fine-tuned model
E-14	science RL without the science fine-tuning	the same RL started from the RL model
long-document RL	long-document variant	RL on inputs of up to 2,048 source tokens
clean rows	rows used for comparisons	benchmark rows minus 214 rows flagged by the contamination audit
cells	language–benchmark pairs	one language on one benchmark; 93 pairs on FLORES+, NTREX and MAFAND-MT; 76 with a held-out AfriCOMET score
TP languages	the 19 languages with extra control data	TranslatePsy languages of the SFT-only control
reward flag, independent flag	the reward's repetition check; the independent repetition detector	see Appendix G
E-01	repetition-penalty control	decoding setting that discourages repeated tokens
E-02	beam search and MBR study	decoding at 768 tokens on 20 languages
E-03	SFT-only control	matched-compute continued SFT
E-04, E-11	seed pair and ablations	extra seeds and feature-removal runs of the first RL stage
E-05	neural scores for all stages	SSA-COMET-QE, AfriCOMET and chrF on AfriScience-MT
E-09, E-09b	engine and precision checks	vLLM against Hugging Face, bf16 against fp32
E-13	earlier 270M models in the same harness	re-run on rows clean for both
E-15, E-16, E-20	long-document, science and B1-final evaluations	evaluation runs
E-10, E-17	human evaluation; NLLB-data ablation	not run
SFT-ctl, RL-A, RL-B1, RL	SFT-only control; first RL stage; second RL stage (best checkpoint); RL model	column labels of the full-results tables (Appendix B); [a] marks the control, which was trained on NLLB-3.3B translations for 19 languages
F+, NT, MF; no_pair; clean	FLORES+, NTREX-128, MAFAND-MT; row sets	*no_pair* = rows whose reference pair the earlier 270M SFT never saw; *clean* = rows whose English side is not in the 270M SFT text
TPsy-0.8B, TPsy-2B	TranslatePsy-AfriSLM models	small translation models for 19 (0.8B) and 9 (2B) of our languages, distilled from NLLB-3.3B outputs

Table 6: AfriScience-MT test, sentence level (six languages, one run per stage). SSA-QE = SSA-COMET-QE, of the reward family for every RL row.

model	chrF	AfriCOMET	SSA-QE
SFT baseline	36.00	0.6064	0.5686
first RL stage	45.06	0.6822	0.6118
RL model (mixed best)	45.40	0.6828	0.6130
science RL without science fine-tuning (E-14)	45.66	0.6850	0.6153
science fine-tuning ($\alpha = 0.5$)	46.63	0.6879	0.6118
science RL final	47.10	0.6932	0.6158
long-document RL final	47.16	0.6936	0.6165

B Full results against other systems

Row sets and how each number was computed. chrF++ is sacreBLEU 2.6.0 `CHRF(word_order=2)` on NFC-normalised text, recomputed from the stored hypotheses for our systems and from stored per-sentence statistics for NLLB and TranslatePsy; corpus statistics are computed per language and then averaged over languages. AfriCOMET is the held-out, reference-based `africomet-stl-1.1` only; the two scoring runs (one for NLLB and TranslatePsy, one for our models) agree to a mean absolute difference of 2.3×10^{-4} on 15,400 identical triples. *no_pair* rows are FLORES+ rows whose reference pair the earlier 270M SFT never saw (42,680 of 44,528 rows); this is the set on which NLLB and TranslatePsy have AfriCOMET scores. *clean* rows are rows whose English side is not in the 270M SFT text (48 sentences per language on FLORES+; 25,221 NTREX and 14,276 MAFAND-MT sentences); only on them are the earlier 270M models not exposure-inflated. NTREX and MAFAND-MT use all rows (our flagged rows, 120 and 94, change the averages by at most 0.006 chrF++). The FLORES+ NLLB set has 44 languages (the published NLLB table has 43; the 44th is Kanuri), 40 of which have an AfriCOMET score in our harness; TranslatePsy-0.8B covers 19 languages and TranslatePsy-2B 9. Intervals are paired bootstrap over test sentences (1,000 resamples, Poisson weights per source sentence shared across systems and, for FLORES+ and NTREX, across languages; seed 20261104). Where NLLB exists only as a per-language corpus-level aggregate (NTREX, MAFAND-MT, SmolSent, SALT) the interval is a bootstrap over languages (10,000 resamples), there is no AfriCOMET, and no sentence pairing is possible. Decoding: ours greedy (768 tokens, vLLM, bf16); NLLB beam search with 4 beams; TranslatePsy greedy with its model-card prompt (256 tokens). The 385M beam-4 run of Section 5.3 covers 20 languages only, so no beam-matched comparison on the full set exists.

Independence and disclosures. chrF++, chrF and spBLEU against human references are independent of every reward. AfriCOMET-STL-1.1 is held out of every reward but belongs to the family of the quality model in the reward and was used to filter the TranslatePsy training rows, so it favours the SFT-only control and TranslatePsy. The SFT-only control was trained on NLLB-3.3B translations (TranslatePsy, CC-BY-NC-4.0) for 19 languages and its checkpoint is not released; TranslatePsy-AfriSLM-0.8B and -2B were distilled from NLLB-3.3B outputs, so their lead over NLLB-3.3B on their languages is a student–teacher effect and not independent evidence for a better model; NLLB models and outputs carry CC-BY-NC terms. The NLLB-derived share of our own SFT data and the NLLB-3.3B pseudo-reference reward term for five languages are listed in Appendix N and in the Ethical considerations. The five languages with that reward term (kea kbp nus bam kab) and the seven with a human-reference reward term are partly optimised towards chrF++. SmolSent, SALT, GlotStoryBook, IbomMT and FLORES-MOZ have no RL model evaluated anywhere. In the tables below SFT = SFT baseline, SFT-ctl = SFT-only control, RL = RL model, RL-A and RL-B1 = first and second RL stage, F+ = FLORES+, NT = NTREX-128, MF = MAFAND-MT, TPsy = TranslatePsy-AfriSLM (Appendix A).

Per-language results

Per-language spBLEU on FLORES+ is in the full version; the language averages of spBLEU for the 44 languages are SFT baseline 7.1, SFT-only control 9.3, RL model 13.2, NLLB-600M 15.1, NLLB-3.3B 17.0 (TranslatePsy-0.8B: 22.4 on its 19 languages).

System	F+ no_pair (44/40)		F+ clean (44/40)		NT all (21/21)		MF all (14/14)	
	chrF++	COMET	chrF++	COMET	chrF++	COMET	chrF++	COMET
SFT	25.6	0.550	22.5	0.468	31.0	0.551	34.5	0.614
SFT-ctl[a]	28.6	0.586	25.3	0.502	35.9	0.609	39.5	0.670
RL-A	34.2	0.620	31.2	0.538	38.6	0.613	39.9	0.656
RL-B1	34.5	0.621	31.8	0.541	38.8	0.613	39.9	0.655
RL	34.9	0.623	32.1	0.544	39.0	0.614	39.9	0.654
270M SFT	–	–	24.9	0.480	–	–	–	–
270M RL	–	–	31.8	0.549	–	–	–	–
NLLB-600M	35.9	0.662	32.5	0.579	–	–	–	–
NLLB-3.3B	**38.1**	**0.677**	**35.0**	**0.601**	**43.3**	–	**44.6**	–

Table 7: All systems, language macro, chrF++ and held-out AfriCOMET-STL-1.1 (per-sentence rows on identical sentences within a panel). RL-A = stage A final, RL-B1 = B1 best, RL = final (mixed best). Bold = best in column; "–" = not available.

Table 11: FLORES+ devtest, chrF++ per language on the no_pair rows (n sentences per language; identical for every system). "–": language not covered by the system. The last rows are macros over the languages with a value, so columns have different language sets (use the main tables for comparisons).

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	63.1	63.3	64.1	61.9	**64.3**	64.2
amh	Amharic	1012	16.6	27.0	26.6	37.3	**38.0**	37.6
bam	Bambara	1012	19.4	20.1	27.2	28.4	**31.0**	–
bem	Bemba	1012	19.3	20.0	33.9	34.7	**37.3**	–
dyu	Dyula	1012	9.9	9.8	**19.0**	16.6	15.4	–
ewe	Ewe	1012	26.0	27.4	31.2	35.2	**38.3**	–
fon	Fon	1012	20.2	19.8	**22.2**	17.9	20.3	–
hau	Hausa	170	35.9	44.5	44.1	45.4	47.3	**48.3**
ibo	Igbo	453	34.5	35.9	37.5	36.9	37.7	**38.8**
kab	Kabyle	1012	16.3	15.9	26.7	27.4	**31.8**	–
kam	Kamba	1012	17.2	16.7	25.9	**26.3**	22.9	–
kbp	Kabiye	1012	14.5	15.5	24.9	23.5	**28.8**	–
kea	Kabuverdianu	1012	18.9	19.0	39.2	44.1	**47.8**	–
kik	Kikuyu	1012	26.0	26.6	33.3	34.7	**37.0**	–
kin	Kinyarwanda	1012	25.8	39.7	38.7	43.5	**48.2**	48.0
kmb	Kimbundu	1012	16.0	16.3	**26.6**	24.8	23.7	–
knc	Kanuri	1012	3.8	3.0	14.2	14.7	**24.4**	–
lin	Lingala	1012	28.6	45.3	44.3	46.3	47.8	**48.8**
lua	Luba-Kasai	1012	17.2	16.0	33.4	33.4	**35.1**	–
lug	Luganda	1010	16.1	34.7	31.7	34.6	38.5	**39.0**
luo	Luo	1012	21.4	21.6	34.5	36.8	**38.4**	–
mlg	Malagasy	1012	39.0	44.2	44.7	47.8	**50.0**	49.6
mos	Mossi	1012	14.1	14.4	22.8	20.7	**23.7**	–
nso	Sepedi	1012	31.6	30.3	49.0	48.9	**50.4**	–
nus	Nuer	1012	13.5	11.4	24.8	25.3	**28.7**	–
nya	Chichewa	1012	36.9	40.2	42.2	44.0	44.8	**45.2**
orm	Oromo	988	12.1	31.3	30.2	31.1	35.6	**37.0**
run	Kirundi	1012	21.6	23.2	33.9	38.9	**41.6**	–
sag	Sango	1012	19.7	19.8	34.3	33.9	**36.4**	–
sna	Shona	1012	33.9	39.2	37.6	42.7	42.6	**43.2**
som	Somali	1012	35.5	38.5	39.7	41.4	42.6	**43.4**
sot	Sesotho	1012	37.9	43.3	42.7	45.4	46.2	**46.7**
ssw	Swati	1012	22.2	22.7	37.9	41.1	**43.1**	–
swh	Swahili	1012	56.2	56.3	58.4	58.0	59.8	**61.1**
tir	Tigrinya	1012	10.5	9.7	18.7	22.4	**24.7**	–
tsn	Setswana	982	37.8	45.6	45.0	46.3	**48.4**	48.3
tso	Tsonga	1012	32.9	29.8	46.9	48.9	**50.1**	–
tum	Tumbuka	1012	25.4	25.3	34.3	33.5	**35.8**	–
twi	Twi	1012	27.5	28.1	35.1	34.9	**38.1**	–
umb	Umbundu	1012	14.8	15.9	**25.5**	25.3	22.9	–
wol	Wolof	1012	26.9	29.1	**30.6**	23.3	27.6	29.8
xho	Xhosa	1012	43.6	44.4	45.4	46.4	48.5	**48.6**
yor	Yoruba	657	19.6	26.2	28.0	25.2	26.2	**28.4**
zul	Zulu	976	46.0	49.6	48.2	51.6	**54.4**	54.2
Macro (languages with a value)			25.6	28.6	34.9	35.9	38.1	45.3
Languages with a value			44	44	44	44	44	19

Table 12: FLORES+ devtest, held-out reference-based AfriCOMET-STL-1.1 per language, same sentences as the chrF++ table (languages outside the AfriCOMET encoder card are "–").

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	0.745	0.754	0.755	0.779	**0.797**	0.787
amh	Amharic	1012	0.504	0.629	0.592	0.712	0.719	**0.751**
bam	Bambara	1012	0.452	0.456	0.521	0.577	**0.596**	–

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
bem	Bemba	1012	0.450	0.455	0.561	0.620	**0.640**	–
dyu	Dyula	1012	–	–	–	–	–	–
ewe	Ewe	1012	0.562	0.571	0.608	0.668	**0.689**	–
fon	Fon	1012	0.573	0.569	**0.603**	0.549	0.573	–
hau	Hausa	170	0.529	0.592	0.568	0.621	0.618	**0.650**
ibo	Igbo	453	0.614	0.649	0.630	0.656	0.658	**0.678**
kab	Kabyle	1012	0.558	0.550	0.674	0.673	**0.708**	–
kam	Kamba	1012	0.440	0.437	0.503	**0.550**	0.528	–
kbp	Kabiye	1012	0.538	0.558	0.661	0.631	**0.679**	–
kea	Kabuverdianu	1012	–	–	–	–	–	–
kik	Kikuyu	1012	0.597	0.606	0.662	0.697	**0.718**	–
kin	Kinyarwanda	1012	0.564	0.687	0.656	0.724	0.746	**0.753**
kmb	Kimbundu	1012	0.419	0.421	0.500	0.506	**0.509**	–
knc	Kanuri	1012	–	–	–	–	–	–
lin	Lingala	1012	0.520	0.642	0.617	0.676	0.684	**0.701**
lua	Luba-Kasai	1012	0.430	0.420	0.556	0.598	**0.615**	–
lug	Luganda	1010	0.464	0.626	0.576	0.635	0.657	**0.671**
luo	Luo	1012	0.562	0.560	0.664	0.712	**0.722**	–
mlg	Malagasy	1012	0.593	0.658	0.634	0.708	**0.727**	0.725
mos	Mossi	1012	0.459	0.459	0.548	0.555	**0.595**	–
nso	Sepedi	1012	0.586	0.588	0.689	0.719	**0.727**	–
nus	Nuer	1012	0.474	0.427	0.624	0.639	**0.681**	–
nya	Chichewa	1012	0.625	0.674	0.651	0.701	0.702	**0.720**
orm	Oromo	988	0.420	0.652	0.610	0.651	0.696	**0.721**
run	Kirundi	1012	0.531	0.552	0.639	0.701	**0.716**	–
sag	Sango	1012	–	–	–	–	–	–
sna	Shona	1012	0.596	0.662	0.619	0.689	0.690	**0.707**
som	Somali	1012	0.590	0.643	0.620	0.669	0.684	**0.702**
sot	Sesotho	1012	0.615	0.670	0.642	0.691	0.695	**0.710**
ssw	Swati	1012	0.493	0.506	0.606	0.681	**0.697**	–
swh	Swahili	1012	0.693	0.718	0.709	0.745	0.753	**0.780**
tir	Tigrinya	1012	0.507	0.487	0.651	0.709	**0.727**	–
tsn	Setswana	982	0.633	0.696	0.674	0.710	0.723	**0.730**
tso	Tsonga	1012	0.597	0.570	0.667	0.715	**0.718**	–
tum	Tumbuka	1012	0.532	0.535	0.605	0.647	**0.673**	–
twi	Twi	1012	0.594	0.600	0.648	0.666	**0.686**	–
umb	Umbundu	1012	0.470	0.474	0.550	**0.580**	0.568	–
wol	Wolof	1012	0.628	0.654	0.653	0.606	0.650	**0.676**
xho	Xhosa	1012	0.627	0.653	0.638	0.686	0.699	**0.703**
yor	Yoruba	657	0.563	0.670	0.682	0.681	0.671	**0.699**
zul	Zulu	976	0.659	0.700	0.674	0.728	0.745	**0.752**
Macro (languages with a value)			0.550	0.586	0.623	0.662	0.677	0.717
Languages with a value			40	40	40	40	40	19

Table 13: FLORES+ devtest, plain chrF (word order 0) per language, same sentences as Table 11.

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	65.2	65.5	66.3	64.1	**66.6**	66.6
amh	Amharic	1012	18.2	29.4	29.0	40.1	**40.8**	40.4
bam	Bambara	1012	20.8	21.6	29.2	30.5	**33.3**	–
bem	Bemba	1012	22.2	23.0	37.9	38.9	**41.6**	–
dyu	Dyula	1012	10.7	10.7	**21.0**	18.5	17.3	–
ewe	Ewe	1012	27.8	29.4	33.4	37.6	**41.1**	–
fon	Fon	1012	21.6	21.2	**23.9**	19.1	21.7	–
hau	Hausa	170	37.7	47.3	46.9	48.1	50.0	**51.1**
ibo	Igbo	453	36.3	37.9	39.6	39.0	39.9	**40.9**
kab	Kabyle	1012	17.4	17.0	28.2	29.0	**33.8**	–
kam	Kamba	1012	19.4	18.8	28.6	**29.1**	25.8	–
kbp	Kabiye	1012	15.8	17.0	26.8	25.2	**30.7**	–
kea	Kabuverdianu	1012	20.2	20.2	40.8	46.3	**50.0**	–
kik	Kikuyu	1012	28.4	29.1	36.1	37.4	**39.9**	–
kin	Kinyarwanda	1012	28.6	43.5	42.6	47.5	**52.3**	52.0
kmb	Kimbundu	1012	18.5	18.9	**30.2**	28.1	26.9	–
knc	Kanuri	1012	4.7	3.8	16.6	16.9	**28.0**	–
lin	Lingala	1012	30.9	48.2	47.2	49.2	50.7	**51.8**
lua	Luba-Kasai	1012	19.7	18.5	37.3	37.4	**39.2**	–
lug	Luganda	1010	18.5	39.1	35.8	39.2	43.5	**43.9**
luo	Luo	1012	23.6	23.9	37.5	39.9	**41.7**	–
mlg	Malagasy	1012	42.4	48.1	48.7	51.8	**54.1**	53.7
mos	Mossi	1012	15.2	15.6	24.6	22.3	**25.6**	–
nso	Sepedi	1012	33.3	32.1	51.3	51.3	**52.8**	–
nus	Nuer	1012	15.0	12.8	27.1	27.5	**31.2**	–
nya	Chichewa	1012	40.7	44.6	46.6	48.6	49.5	**49.9**
orm	Oromo	988	14.4	36.3	35.2	36.4	41.2	**42.8**
run	Kirundi	1012	24.5	26.4	38.2	43.4	**46.2**	–
sag	Sango	1012	20.7	20.9	36.3	35.8	**38.5**	–
sna	Shona	1012	38.4	44.4	42.7	48.0	48.1	**48.7**
som	Somali	1012	39.4	42.6	44.0	45.7	47.1	**47.8**
sot	Sesotho	1012	40.4	46.2	45.5	48.3	49.3	**49.7**
ssw	Swati	1012	26.0	26.7	43.8	47.1	**49.2**	–
swh	Swahili	1012	58.6	58.8	60.8	60.5	62.4	**63.6**
tir	Tigrinya	1012	11.6	10.8	20.6	24.3	**26.6**	–
tsn	Setswana	982	40.0	48.1	47.5	48.9	**50.9**	50.8
tso	Tsonga	1012	35.5	32.4	50.1	52.1	**53.3**	–
tum	Tumbuka	1012	28.4	28.4	38.0	36.8	**39.2**	–
twi	Twi	1012	29.2	29.9	37.0	36.9	**40.1**	–

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
umb	Umbundu	1012	17.8	19.2	**30.1**	29.7	27.1	–
wol	Wolof	1012	28.5	30.9	**32.4**	24.7	29.4	31.6
xho	Xhosa	1012	49.2	50.3	51.4	52.5	54.6	**54.7**
yor	Yoruba	657	21.3	27.9	29.7	26.9	28.2	**30.3**
zul	Zulu	976	51.3	55.1	53.8	57.2	**60.0**	59.8
Macro (languages with a value)			27.9	31.2	37.9	39.0	41.4	48.9
Languages with a value			44	44	44	44	44	19

Table 14: NTREX-128 per language, all rows (21 languages): chrF++ and held-out AfriCOMET-STL-1.1. NLLB-3.3B is a corpus-level aggregate on the same rows (no AfriCOMET exists for it). SFT-ctl[a] = SFT-only control, trained on NLLB-3.3B translations for 19 languages and never released.

		chrF++				AfriCOMET		
ISO	Language	SFT	SFT-ctl[a]	RL	NLLB-3.3B	SFT	SFT-ctl[a]	RL
afr	Afrikaans	62.4	63.6	63.9	**68.5**	0.682	**0.703**	0.698
amh	Amharic	10.6	18.1	18.0	**25.4**	0.444	**0.558**	0.521
bem	Bemba	18.7	18.8	34.8	**38.6**	0.433	0.436	**0.544**
ewe	Ewe	24.3	25.7	31.6	**36.9**	0.519	0.535	**0.583**
hau	Hausa	42.1	49.5	48.6	**52.9**	0.597	**0.662**	0.625
ibo	Igbo	39.9	43.3	44.9	**47.1**	0.645	**0.689**	0.673
kin	Kinyarwanda	26.4	38.4	37.1	**45.8**	0.551	**0.660**	0.623
mlg	Malagasy	36.9	41.5	40.8	**46.6**	0.547	**0.613**	0.573
nso	Sepedi	20.1	19.7	32.7	**35.0**	0.487	0.497	**0.588**
nya	Chichewa	37.9	43.2	44.2	**48.7**	0.588	**0.655**	0.614
orm	Oromo	9.6	27.3	27.3	**31.2**	0.395	**0.595**	0.562
sna	Shona	34.2	40.6	39.2	**45.1**	0.573	**0.644**	0.597
som	Somali	35.8	40.6	44.2	**47.7**	0.549	**0.623**	0.610
ssw	Swati	21.6	20.9	38.8	**45.3**	0.471	0.474	**0.593**
swh	Swahili	55.2	57.6	58.7	**61.8**	0.646	**0.679**	0.664
tir	Tigrinya	9.1	9.8	17.0	**23.8**	0.488	0.493	**0.632**
tsn	Setswana	37.0	49.4	48.7	**54.8**	0.577	**0.683**	0.650
wol	Wolof	25.3	29.5	**30.4**	29.4	0.593	**0.635**	0.625
xho	Xhosa	39.8	40.7	42.4	**43.8**	0.584	**0.618**	0.598
yor	Yoruba	23.7	30.9	32.1	**32.1**	0.597	0.691	**0.701**
zul	Zulu	40.9	44.1	43.5	**47.9**	0.600	**0.643**	0.616
Macro (languages with a value)		31.0	35.9	39.0	43.3	0.551	0.609	0.614
Languages with a value		21	21	21	21	21	21	21

Table 15: MAFAND-MT per language, all rows (14 languages): chrF++ and held-out AfriCOMET-STL-1.1. NLLB-3.3B is a corpus-level aggregate on the same rows (no AfriCOMET exists for it).

		chrF++				AfriCOMET		
ISO	Language	SFT	SFT-ctl[a]	RL	NLLB-3.3B	SFT	SFT-ctl[a]	RL
amh	Amharic	11.6	18.5	18.3	**23.7**	0.456	**0.556**	0.515
hau	Hausa	32.0	33.7	32.8	**35.3**	0.630	**0.671**	0.644
ibo	Igbo	43.6	44.7	44.9	**48.4**	0.700	**0.720**	0.706
kin	Kinyarwanda	36.5	48.1	46.4	**56.3**	0.623	**0.716**	0.688
lug	Luganda	27.0	45.4	39.6	**52.6**	0.558	**0.680**	0.623
luo	Luo	23.0	25.0	32.6	**39.7**	0.579	0.587	**0.651**
nya	Chichewa	38.1	42.1	41.0	**43.3**	0.627	**0.681**	0.636
sna	Shona	30.4	34.3	32.9	**38.1**	0.583	**0.648**	0.601
swh	Swahili	50.6	52.0	51.8	**55.3**	0.680	**0.711**	0.693
tsn	Setswana	43.9	51.0	53.1	**56.3**	0.667	**0.716**	0.712
twi	Twi	28.6	29.1	34.1	**36.5**	0.601	0.599	**0.643**
xho	Xhosa	49.8	51.5	51.8	**56.7**	0.672	**0.706**	0.680
yor	Yoruba	23.5	32.1	**33.4**	32.7	0.614	0.725	**0.740**
zul	Zulu	44.5	46.4	45.6	**49.5**	0.610	**0.657**	0.620
Macro (languages with a value)		34.5	39.5	39.9	44.6	0.614	0.670	0.654
Languages with a value		14	14	14	14	14	14	14

Row set (langs)	RL − system	ΔchrF++	95% CI	ΔCOMET	95% CI
F+ no_pair (44/40)	SFT	9.30	[9.00, 9.55]	0.0732	[0.0711, 0.0752]
	SFT-ctl[a]	6.32	[6.11, 6.52]	0.0377	[0.0360, 0.0394]
	RL-A	0.72	[0.65, 0.81]	0.0039	[0.0031, 0.0046]
	RL-B1	0.34	[0.29, 0.41]	0.0024	[0.0017, 0.0030]
	NLLB-600M	-1.05	[-1.24, -0.86]	-0.0381	[-0.0399, -0.0362]
	NLLB-3.3B	-3.21	[-3.38, -3.02]	-0.0535	[-0.0553, -0.0518]
F+ TPsy-19 (19/19)	SFT	7.04	[6.64, 7.41]	0.0536	[0.0513, 0.0559]
	SFT-ctl[a]	0.08	[-0.13, 0.30]	-0.0226	[-0.0242, -0.0208]
	RL-A	0.29	[0.14, 0.43]	0.0008	[-0.0004, 0.0018]
	RL-B1	0.14	[0.04, 0.26]	0.0010	[-0.0000, 0.0020]
	NLLB-600M	-1.54	[-1.78, -1.28]	-0.0456	[-0.0480, -0.0434]
	NLLB-3.3B	-3.62	[-3.84, -3.39]	-0.0586	[-0.0609, -0.0563]
	TPsy-0.8B	-4.23	[-4.39, -4.07]	-0.0745	[-0.0765, -0.0727]
NT all (21/21)	SFT	7.97	[7.73, 8.21]	0.0630	[0.0614, 0.0646]
	SFT-ctl[a]	3.14	[2.97, 3.32]	0.0050	[0.0037, 0.0063]
	RL-A	0.41	[0.34, 0.48]	0.0013	[0.0006, 0.0020]
	RL-B1	0.16	[0.10, 0.23]	0.0004	[-0.0002, 0.0010]
	NLLB-3.3B	-4.26	[-5.22, -3.24][ℓ]	–	–
MF all (14/14)	SFT	5.39	[5.15, 5.63]	0.0395	[0.0377, 0.0413]
	SFT-ctl[a]	0.34	[0.16, 0.52]	-0.0158	[-0.0174, -0.0143]
	RL-A	-0.05	[-0.13, 0.05]	-0.0025	[-0.0035, -0.0015]
	RL-B1	-0.04	[-0.11, 0.03]	-0.0012	[-0.0021, -0.0003]
	NLLB-3.3B	-4.73	[-6.56, -3.12][ℓ]	–	–
F+ clean (44/40)	SFT	9.61	[8.34, 10.62]	0.0752	[0.0670, 0.0836]
	SFT-ctl[a]	6.87	[5.98, 7.61]	0.0416	[0.0335, 0.0495]
	RL-A	0.90	[0.58, 1.20]	0.0055	[0.0020, 0.0093]
	RL-B1	0.33	[0.10, 0.58]	0.0026	[-0.0006, 0.0057]
	270M SFT	7.20	[6.31, 7.94]	0.0639	[0.0574, 0.0709]
	270M RL	0.35	[0.08, 0.63]	-0.0052	[-0.0102, -0.0003]
	NLLB-600M	-0.39	[-1.13, 0.51]	-0.0353	[-0.0423, -0.0275]
	NLLB-3.3B	-2.85	[-3.57, -1.99]	-0.0576	[-0.0655, -0.0503]

Table 8: Final RL model minus each other system, language-macro difference with 95% paired bootstrap interval (1,000 resamples, Poisson weights per source sentence shared across systems and, for FLORES+ and NTREX, across languages; seed 20261104). Positive = RL better. Counts: languages for chrF++ / COMET. [ℓ]: no per-sentence output of NLLB-3.3B on NTREX/MAFAND is archived; the interval is a bootstrap over languages (language sampling only). SFT-ctl[a]: see Table 1.

Table 9: SSA-COMET-MTL (reference-based, $\times 100$) on FLORES+ *no_pair* rows. **Reward family: not independent evidence.** No value exists for any 385M model; the 270M rows come from the earlier work's own run, not the re-run (the re-run of the earlier models is in the FLORES+ clean columns of Table 7).

system	languages	own languages	19 TPsy-0.8B languages	9 TPsy-2B languages
NLLB-600M	22	56.71	56.72	56.91
NLLB-3.3B	22	59.28	59.14	58.90
TranslatePsy-0.8B	19	60.75	60.75	60.31
TranslatePsy-2B	9	60.43	–	60.43
earlier 270M RL-final	22	51.92	52.81	50.29
earlier 270M SFT-final	22	47.54	48.78	46.19

Table 10: Against published whole-devtest NLLB numbers (FLORES-200 devtest chrF++; our systems rescored on all 1,012 FLORES+ rows; 43 shared languages; *not* sentence-paired; intervals over languages, 10,000 resamples).

system	chrF++	RL − system
RL model	35.54	
SFT baseline / SFT-only control / RL-A / RL-B1	26.30 / 29.35 / 34.91 / 35.25	$+9.24$ / $+6.19$ / $+0.63$ / $+0.29$
NLLB-600M, our rerun	36.62	-1.08 $[-1.98, -0.20]$
NLLB-3.3B, our rerun	38.63	-3.09 $[-4.13, -2.05]$
NLLB-3.3B, published	38.80	-3.26 $[-4.30, -2.21]$
NLLB-1.3B, published	38.09	-2.55 $[-3.46, -1.62]$
NLLB-distilled-1.3B, published	38.34	-2.80 $[-3.72, -1.87]$
NLLB-MoE-54.5B, published	39.08	-3.54 $[-4.59, -2.52]$

ISO	Language	n	SFT	NLLB-3.3B
SmolSent (no RL model evaluated)				
amh	Amharic	863	10.3	**30.6**
bam	Bambara	863	13.3	**26.5**
bem	Bemba	863	13.8	**32.3**
dyu	Dyula	863	10.5	**20.8**
ewe	Ewe	863	20.6	**36.0**
fon	Fon	863	15.5	**17.8**
hau	Hausa	863	28.2	**40.4**
ibo	Igbo	863	35.7	**48.4**
kab	Kabyle	862	11.6	**27.8**
kik	Kikuyu	863	21.2	**40.1**
kin	Kinyarwanda	863	18.8	**41.8**
kon	Kongo	863	22.4	**47.8**
lin	Lingala	863	22.5	**45.1**
lug	Luganda	863	14.6	**39.4**
luo	Luo	863	16.0	**32.7**
mlg	Malagasy	863	31.6	**48.1**
mos	Mossi	863	11.5	**18.8**
nso	Sepedi	863	22.5	**45.0**
nya	Chichewa	863	25.6	**38.0**
orm	Oromo	863	8.5	**34.9**
run	Kirundi	863	15.1	**33.7**
sna	Shona	863	27.5	**40.2**
som	Somali	862	23.5	**35.4**
sot	Sesotho	863	25.9	**38.9**
ssw	Swati	863	18.2	**38.8**
swh	Swahili	863	48.6	**52.4**
tir	Tigrinya	863	6.2	**25.3**
tsn	Setswana	863	34.3	**50.4**
tso	Tsonga	863	29.9	**43.7**
tum	Tumbuka	863	19.5	**38.8**
twi	Twi	863	20.5	**31.3**
wol	Wolof	863	21.3	**23.3**
xho	Xhosa	863	36.6	**43.0**
yor	Yoruba	863	16.6	**28.6**
zul	Zulu	863	40.6	**51.3**
Macro (languages with a value)			21.7	36.8
Languages with a value			35	35
SALT (no RL model evaluated)				
ibo	Igbo	500	32.4	**41.5**
lug	Luganda	500	29.3	**52.1**
swh	Swahili	500	56.8	**61.9**
Macro (languages with a value)			39.5	51.8
Languages with a value			3	3

Table 16: SmolSent and SALT, chrF++ per language, all rows: SFT baseline and NLLB-3.3B (corpus-level aggregate). No RL model was evaluated on these benchmarks.

C Hyper-parameters, recipe and reward details

C.1 SFT recipe and learning-rate schedule

Recipe. Samples are packed into 4,096-token packs (one language per pack, block-diagonal attention) at 32 packs = 131,072 tokens per step; each pack is used once. Optimiser: Muon (momentum 0.95, 5 Newton-Schulz steps) on the 2-D decoder matrices and AdamW (betas 0.9/0.95) on embeddings and norms, weight decay 0.01, gradient clipping 1.0, 3% linear warm-up then cosine decay, seed 1234. The learning-rate schedule, read from the executed per-shard configs, is: SFT-1 (`g1a`, 1,500 steps) peak 5e-5, cosine to 50% of the peak (2.5e-5); shard 1 (`g1b_s1`) peak 5e-5, cosine to 80% (4e-5); shards 2-5 (`g1b_s2`..`g1b_s5`) peak 3e-5 (lowered from 5e-5 on 2026-11-01 because of forgetting in tiny languages), cosine to 80% (2.4e-5); shard 6 (`g1b_s6`) peak 3e-5, cosine to 10%, so the last step runs at 3.0e-6. Every shard starts a fresh Muon state with a 3% linear warm-up. SFT-2 runs in six shards of about 4,800 to 5,300 steps, with tail-language replay from shard 2. Executed stage B: 31,353 steps, 4.109 B tokens, 5.14 s per step on an A100 (44.8 h); 32,853 steps in all with SFT-1. Final validation loss: macro 1.3412, micro 1.2846 over 78 languages (end of SFT-1: 1.7847). The macro rises from 1.78 to 1.98 at the end of shard 1 because new languages enter the validation set at stage 2.

C.2 RL: stages, reward and features

Reward, the four added features, the learning-rate history of the first RL stage and the monitors are described in Section 3 and in the supplementary material.

The science SFT before the science RL used AdamW at 3e-6 (Appendix K). Dynamic-sampling, std-floor and reference-domain settings differ by stage as well (Appendix E).

Stages. Stage A (400 steps, short English windows, bin b64), B1 (135 steps run, b256; best checkpoint step 50 by dev chrF++), mixed (60 steps, b64 and b256), then a continued-RL run on eight weak languages (85 steps), a science SFT with weight interpolation, a science RL and a long-document RL (Appendix E). B1 started from stage A's final model (step 400), mixed from B1's best checkpoint (step 50), the weak-group run from the mixed best, the science SFT from the mixed best. The dev set has **720 sentences = 18 FLORES+ dev languages** $\times$ **40 sentences** (afr ewe fon hau ibo kin nya orm sna som sot swh tsn twi wol xho yor zul). B1's step-135 weights (Hub branch `stage-b1-final`) were evaluated after the fact (E-20, Section 5.5) and equal B1 best within noise.

Table 17: RL hyper-parameters by stage (all Muon; KL reference = start model of the stage).

stage	learning rate	KL coefficient (k3)	group size	guide member	steps (cap) × prompts
A, B1, mixed	2e-6 (10 warm-up steps)	0.05	16 (15 + greedy)	beam/MBR, fraction 0.25 (from step 41 in A)	400 × 64; 150 (135 run) × 48; 60 × 42
science RL	1.0e-6	0.1	16	beam/MBR, fraction 0.25	150 in the yaml, cap extended to 190 by the extension rule at step 120; executed 160 (stopped at step 150 by dev plateau, then 10 wind-down steps; `summary.json`, Appendix E) × 42
long-document RL	1.0e-6	0.1	8 (7 + greedy)	off	80 × 32 (51% long prompts)

D Earlier 270M model and growth

The comparison of the 385M SFT baseline and RL model with the earlier 270M models in the same harness is in the clean-row columns of Table 7 (Appendix B); per-benchmark comparisons with the pre-growth model A0 and the NLLB-mined-data analysis are in the supplementary material released on acceptance.

E Stage chain and B1 final

The stage chain (first, second and third RL stage, weak-group RL, science and long-document variants) with steps, prompts and GPU cost is in the compute table of Appendix Q and in the supplementary material; the second RL stage evaluated at its final step (step 135) equals its best checkpoint (step 50) within noise (+0.01 chrF++, 93 language–benchmark pairs).

F RL against SFT and the control: tables

Language-macro chrF++ per benchmark, SFT / control / mixed best: FLORES+ 22.85 / 25.38 / 31.66, NTREX 29.25 / 33.74 / 37.17, MAFAND 35.77 / 40.44 / 40.78; the control gains +2.53 [+2.35, +2.71], +4.49 [+4.29, +4.70] and +4.67 [+4.39, +4.92] over the SFT against +8.81, +7.91 and +5.01 for the RL; over the 93 cells it gains +3.38 against +7.96 (42.5% of the SFT-to-RL gap, arithmetic of the macros). Held-out AfriCOMET: RL minus control is +0.0373 on FLORES+, +0.0051 on NTREX and −0.0141 on MAFAND (the last favours the control). Finished and loop rates (the reward's flag) on FLORES+, SFT / control / mixed best: 0.829 / 0.854 / 0.995 and 0.209 / 0.178 / 0.048; by panel (FLORES+ finished): the 19 TranslatePsy languages 0.931 / 0.993 / 1.000, the other languages 0.774 / 0.779 / 0.993. The independent degeneration detector gives the same picture (FLORES+ flag 0.311 for the SFT, 0.268 for the control, 0.085 for the mixed best). **The control repairs finishing and looping only on the languages it was trained on**, where it comes close to the RL model; on the other languages it changes almost nothing (+0.15 chrF++ over the SFT on FLORES+ [−0.03, +0.32], 35 languages).

Table 18: The pre-registered decision rule applied to mixed best against the 6,000-step control.

registered condition	value	met
lower 95% bound of d (chrF++, 93 cells) $> +1.0$	$+4.58$ [$+4.45$, $+4.70$]	yes
lower bound of the AfriCOMET difference > 0	$+0.0183$ [$+0.0172$, $+0.0192$]	yes
$d > 0$ on the 19 TranslatePsy languages	-0.03 [-0.13, $+0.09$]	**no**
$d > 0$ on the other languages	$+9.29$ [$+9.09$, $+9.49$]	yes

G Independent degeneration audit: tables

- **What the audit does not establish.** There is no ground truth for "degenerate"; the detector is validated by the reference control, by reading 25 mixed-best and 12 SFT examples, and by the lower language-matched chrF++. It is blind to non-repetitive failures (hallucination, wrong content) except through the length audit. The NLLB LID is licensed cc-by-nc-4.0 and was used for evaluation only.

Table 19: Loop flag of the reward against the independent detector, language macro (95% bootstrap in brackets).

system	FLORES+ reward flag	FLORES+ independent flag	NTREX reward / independent	MAFAND reward / independent
SFT	0.2095	0.3107 [0.3029, 0.3190]	0.1218 / 0.1662	0.0431 / 0.0754
RL stage A	0.0675	0.1207 [0.1160, 0.1253]	0.0165 / 0.0309	0.0026 / 0.0092
RL B1 best	0.0541	0.0980 [0.0942, 0.1019]	0.0092 / 0.0190	0.0013 / 0.0056
RL mixed best	0.0480 [0.0462, 0.0496]	0.0851 [0.0822, 0.0883]	0.0064 / 0.0143	0.0009 / 0.0039
human references	0.0001	0.0018	0.0000 / 0.0018	0.0000 / 0.0013

H Seeds and ablations (E-04, E-11): full table

Seed spread (E-04). Two extra runs of the stage-A recipe (seeds 20261105 and 20261106; 100 steps; the executed stage A used seed 1234, so the three runs share the recipe, the SFT start and the 40-step feature-free prefix schedule) were evaluated with the standard harness on 20 languages (yor hau ibo swh amh zul xho kin som wol lug sna twi ewe lin tir orm nso tsn afr; FLORES+, NTREX, MAFAND; 49 cells; greedy, 768 tokens). Gains over the SFT are +5.96 [+5.80, +6.11] and +5.90 [+5.74, +6.05] chrF++ (+0.0462 and +0.0454 AfriCOMET): the two seeds differ by **0.06 chrF++ [−0.02, +0.14]** (0.0008 AfriCOMET; per language they differ significantly in 2 of 20, by at most 0.54). The in-training dev gains of the three runs are +3.33, +3.70 and +3.46 (the dev subset depends on the seed, so only gains over each run's own step 0 are compared). The stage-A gain is therefore reproducible across training seeds at step 100, and per-language differences below about 0.5 chrF++ are not separable from training noise. **This is one pair of seeds at step 100, not a seed study of later stages**; the 400-step stage A is +0.78 [+0.70, +0.86] and +0.84 [+0.75, +0.92] above the 100-step seeds on the same cells.

Removing all four features costs **0.32 chrF++ [0.23, 0.41]** (about five times the seed difference; consistent on the three benchmarks, AfriCOMET and the loop flag), about 5% of the 5.96 chrF++ gain, and it does not show on the in-training dev macro (−0.14 [−0.51, +0.25]). Removing the reference domain costs 0.16 [0.08, 0.25] (real by the rule against seed 2, 2.7 times the seed difference): about half of the feature set's effect. Removing the beam guide costs 0.09 against seed 2 but 0.04 [−0.04, +0.11] against seed 3 with AfriCOMET inside noise: **not separable from seed noise**, for about +25% step time (median step 40.3 s with the guide against 32.4 s without). Dynamic sampling and the std floor never acted and are part of arm (a) only. **The KL rise is not caused by the features:** the median KL of steps 81 to 90 is 0.116 without any feature, 0.122 without the reference domain, 0.120 without the guide and 0.123 and 0.120 for the two full runs, and the same holds at steps 51 to 60 (0.075, 0.0745, 0.073, 0.072, 0.073). **The earlier inference that the beam guide drives the KL slope (stage report, ledger M10) is therefore not supported**; the step-41 acceleration seen in stage A is at most a small effect. Arm (e) (KL guard 0.25 against 0.15) was not run by the pre-registered rule, because the guard never fired in any of the 100-step runs (maximum 5-step median KL 0.120 to 0.127 against 0.15). Caveats: one run per arm, a shared 40-step prefix with seed 2, one pair of seeds as the noise yardstick (so the "real" rule is lenient and effects of 0.1 to 0.2 chrF++ are small and suggestive), 100 steps and 20 languages only (the reference domain covers amh efi hau lug nso yor zul, five of them in these 20). **Reading:** the four features are design choices with a small measured benefit (+0.3 chrF++ at 100 steps, mostly the reference-anchored term), not the source of the RL gain, and the beam guide is not shown to be worth its generation time.

Table 20: Ablation arms at 100 steps, chrF++ over 49 cells (20 languages); arm minus the full-feature seed 2 (paired) and minus seed 3 (E-11).

arm	chrF++ (49 cells)	arm minus seed 2	arm minus seed 3	AfriCOMET vs seed 2
(b) full, seed 2	38.51 [38.30, 38.69]	-	-	-
(b) full, seed 3	38.46 [38.26, 38.64]	-	-	-
(a) no features	38.19 [37.99, 38.38]	−0.32 [−0.41, −0.23]	−0.26 [−0.35, −0.18]	−0.0028 [−0.0033, −0.0022]
(c) no reference domain	38.35 [38.16, 38.54]	−0.16 [−0.25, −0.08]	−0.10 [−0.19, −0.02]	−0.0017 [−0.0023, −0.0011]
(d) no beam guide	38.42 [38.23, 38.61]	−0.09 [−0.17, −0.01]	−0.04 [−0.11, +0.04]	−0.0005 [−0.0010, +0.0001]

I Decoding and engine parity: tables

The decode-time frequency-penalty guard was run on both models over the full evaluation set (E-01, Section 5.3). Engine parity: vLLM bf16 and HF bf16 give identical outputs in 9.6% of 240 sentences (chrF++ 36.22 and 36.80; HF fp32 38.18); in fp32 the two engines agree (98.8% identical for the SFT, 97.1% for the RL model; E-09), so the divergence is bf16 rounding. **Precision does not move the headline:** the SFT in fp32 on the full evaluation set (vLLM, greedy, 768 tokens, 122,585 rows, 214 flagged rows removed; E-09b) scores −0.04 chrF++ [−0.13, +0.07] against bf16 over the 93 cells (FLORES+ −0.16 [−0.29, −0.02], NTREX +0.08 [−0.08, +0.25], MAFAND +0.22 [−0.03, +0.50]; plain chrF −0.08 [−0.19, +0.03]), and the RL gain on FLORES+ is +8.97 [+8.71, +9.20] over the fp32 SFT against +8.81 [+8.59, +9.03] over the bf16 SFT, so the bf16 harness does not overstate the RL gain (the upper end of the all-cell interval bounds any understatement of the SFT at 0.07 chrF++). The +1.81 [+0.59, +3.12] of the 240-sentence set was a small-sample result and the earlier +0.24 [−0.07, +0.55] (4,000 sentences) to +1.81 range for the SFT is superseded. Not done: an fp32 run of the RL model (its sensitivity is measured only on the 240-sentence set, +0.09 [−0.41, +0.59]) and AfriCOMET scoring of the fp32 SFT run (the run was stopped before the scoring finished; the partial scores are not reported).

Table 21: E-02: greedy, beam-4 and MBR-16 at 768 new tokens, full table (20 languages, 4,000 sentences per system, 95% paired bootstrap).

system	chrF++	spBLEU	AfriCOMET-STL	finished	loop (independent)	output / reference length
SFT greedy	32.31 [31.43, 33.38]	11.47	0.5513 [0.5403, 0.5621]	0.915	0.147	2.05
SFT beam-4 (stop at 4 ended)	36.02 [35.13, 36.95]	16.49	0.5639 [0.5521, 0.5750]	1.000	0.108	0.96
SFT beam-4 (HF early_stopping=False)	33.26 [32.22, 34.46]	13.07	0.5722 [0.5603, 0.5845]	0.897	0.139	2.15
SFT MBR-16 (metric-matched)	38.27 [37.60, 39.01]	15.25	0.5780 [0.5687, 0.5870]	0.9998	0.016	1.16
RL greedy	39.87 [39.17, 40.62]	17.39	0.6082 [0.5991, 0.6172]	1.000	0.007	1.05
RL beam-4 (stop at 4 ended)	39.10 [38.20, 39.96]	17.83	0.6045 [0.5944, 0.6148]	1.000	0.007	0.98
RL beam-4 (HF early_stopping=False)	40.70 [40.00, 41.42]	18.47	0.6227 [0.6139, 0.6317]	0.9998	0.007	1.03
RL MBR-16 (metric-matched)	40.79 [40.09, 41.54]	17.74	0.6168 [0.6079, 0.6258]	1.000	0.003	1.08

J NLLB comparison: same sentences and AfriqueLLM family

Earlier same-sentence rows (post-editor run, superseded). NLLB-3.3B and RL stage A drafts were scored on 100-sentence FLORES+ subsets (language bootstrap only): stage A with beam-4 (the beam setup of the post-editor run, not that of E-02) against NLLB greedy was 1.44 below NLLB-3.3B on 43 languages (36.12 against 37.56, $[-2.34, -0.56]$). That run's NLLB output was 1.23 $[-1.60, -0.86]$ below the published full-devtest value on its 100 sentences; on all 1,012 rows our NLLB-3.3B run is only 0.17 below the published value (38.63 against 38.80), so the 1.23 was sample noise, not a systematic effect of FLORES+ against FLORES-200 references. The sentence-paired comparison above on all rows replaces these numbers. AfriCOMET is the QE reward's family, so AfriCOMET rows are not independent evidence for the RL model.

AfriqueLLM family, indicative only. On the 21 languages with per-language values in the 5-shot AfroBench-Lite harness (afr amh ewe hau ibo kin lin lug nya orm sna som sot swh tir tsn twi wol xho yor zul): SFT 32.37, stage A 39.13, B1 best 39.32, mixed best 39.40, weak final 39.26, against AfriqueQwen3.5-4B-50Langs 41.71 (mixed best -2.31; ahead in 4 of 21 languages), AfriqueGemma-12B 38.58 (+0.83; 6 of 21), AfriqueLlama-8B 36.71, AfriqueQwen-14B 36.67, AfriqueQwen-8B 35.06, AfriqueGemma-4B 31.12, Gemma3-12B 18.82, Llama3.1-8B 16.45. Those are base models in 5-shot prompting with a harness whose chrF++ implementation and FLORES split are not verified. Bidirectional aggregates (TranslatePsy, AfriNLLB) and pooled AfroBench rows are not comparable and are not compared.

K Science and document results: tables

Science RL without the science SFT (E-14). To ask whether the science SFT is needed, the executed science RL was repeated with the start model changed to the mixed best (= science-SFT alpha 0, the model the science SFT was built from): same seed 1234, same pool, same reward, Muon at 1e-6, KL 0.1 against the start model, 160 steps (150 plus a 10-step wind-down), pre-declared checkpoint = final. The rule was registered before the run: the science SFT is *not needed* if the lower 95% bounds of E-14 minus the science RL final on AfriScience-MT test are above -0.5 chrF and above -0.004 held-out AfriCOMET; it is *needed* if the upper bound of either is below its margin.

Table 22: E-14 final minus X, 95% bootstrap (independent weights per language).

E-14 final minus	chrF	AfriCOMET-STL (held out)	SSA-COMET-QE (reward family)
science RL final (from the science SFT)	**-1.44 [−1.61, −1.25]**	**-0.0082 [−0.0101, −0.0065]**	−0.0006 [−0.0018, +0.0007]
science SFT alpha 0.5	−0.97 [−1.20, −0.75]	−0.0030 [−0.0050, −0.0009]	+0.0035 [+0.0022, +0.0047]
mixed best (its own start)	+0.26 [+0.12, +0.39]	+0.0022 [+0.0007, +0.0036]	+0.0023 [+0.0013, +0.0033]
B1 final	+0.21 [+0.07, +0.36]	+0.0004 [−0.0012, +0.0020]	+0.0012 [+0.0000, +0.0023]

L Analyses condensed in this version

The following analyses are summarised in the main text and stated here only as pointers. Details are in the supplementary material released on acceptance. **Stage-A log analysis:** the stage-A log cannot separate the four features; the ablations of Section 5.5 can. **Family, script and resource level:** an exploratory, observational analysis over 54 languages whose main result changes with the languages included; it is not reported as a finding. **NLLB-mined data and the regression against the earlier 270M model:** inconclusive. **Negative results** (weak-group RL, science SFT on the headline metric, a post-editor, whole-document decoding) are in Sections 5.5 and 5.6. **Contamination follow-ups** beyond the residual counts listed in Ethical considerations (scan coverage, candidate lists) and the extended limitations list are also in the supplementary material.

M Metric independence: full table

Table 23: Metric independence (full). IND = not in any reward; RWD = in the reward; LOOP = optimised by construction.

metric	in the RL reward?	use in this paper
`masakhane/africomet-qe-stl-1.1` (reference-free)	yes	not used as evidence
`McGill-NLP/ssa-comet-mtl` (reference-free; reference-based in the science RL)	yes	not used as evidence of RL gains
`McGill-NLP/ssa-comet-qe` (AfriScience-MT headline)	no, but the same family as the reward member	headline for SFT-only comparisons; for RL models shown only next to plain chrF and held-out AfriCOMET (all stages in Appendix K)
TranslatePsy quality filters (AfriCOMET, SSA-COMET, MetricX)	not a reward; they filtered the training data of the E-03 control	neural metrics favour the control; chrF and chrF++ are the evidence (Section 5.4)
consensus utility (sample agreement in chrF++)	yes	chrF++ gains are partly in the optimised family
NLLB-3.3B pseudo-reference chrF++	yes, 5 languages (kea kbp nus bam kab)	chrF++ on those 5 languages is not independent; relevant to the NLLB comparison (Section 5.5)
chrF++ against a human reference, weight 0.35	yes, 7 languages (reference rows only; AfriScience TRAIN pairs)	chrF++ on those 7 languages can be partly trained; the test sets differ
language-ID, copy, repetition, length, end-of-turn guards	yes	finished and loop rates are optimised by construction
independent degeneration detector (lzma residual, repeated character 8-grams, repeated word 3-grams; thresholds from the 0.1% tail of human references)	no	measures the same idea as the loop guard with different code and constants (Section 5.3)
NLLB-218 language identification (`facebook/fasttext-language-identification`)	no	independent check of the language guard; evaluation use only (cc-by-nc-4.0)
`masakhane/africomet-stl-1.1` (reference-based)	no (held out; monitor and alarm)	the independent neural check; same family as the QE member
chrF / chrF++ / spBLEU against human references of the benchmarks	no	the main lexical evidence

N Disclosures: full statement

Rule: data and signals that were actually used are stated, including those that are non-commercial or NLLB-derived. `DISCLOSURES.md` is the source of this section and holds the full tables.

N.1 Training data and licences

- **NLLB-derived share of the SFT.** NLLB-mined bitext (OPUS-NLLB, 43 languages, 13.46 M rows, plus a Fulah NLLB-mined set of 0.45 M rows) is **14.4% of the 3.990 B stage-B pack tokens** (13.3% of the 4.109 B executed; 10.84% of the 5.043 B tokens drawn in the whole SFT); adding NLLB-Seed (tzm) and the aligned part of the Aya collection (NLLB-3.3B outputs; 169 M tokens in 12 languages) gives **18.7%** (746 M tokens; executed 17.2%; 14.36% of the whole SFT); adding the machine-translated Tumbuka instruction sets (system not stated, possibly NLLB) gives **22.8%** (909 M tokens; 17.86% of the whole SFT). By rows, 13.91 M of 42.31 M stage-B translation pairs (32.9%) are NLLB-mined. A0 contains none of the three. The authors allowed NLLB-mined rows with a per-language cap of 150k, ODC-BY with special attribution. NLLB models and outputs carry CC-BY-NC terms and the authors cannot use NLLB for production translation; a licence statement must accompany any release of the 385M weights. Not counted and needing a legal decision: the `*_mt560` OPUS sets (69% of the main pair output of the clean rebuild, 16.2 M of 23.4 M pairs, mostly Jehovah's Witnesses and Bible text, licence unknown, labelled NLLB-derived by an old converter docstring); `African-Languages-Lab/multi-open` (3,748,964 pairs, "license: other"); AfroLingu-MT (gated, authors' exception); other machine-translated instruction sets with unstated systems (`african_ultrachat`, `kiya_sft`, `alpaca_*`, `amharic_*`); all four families are in the instruction pack stream (`packed_instr`: every source of the instruction intake except `muri_it`, 4,190,976 samples, 1.232 B tokens; `reports/intake2_instr_packs_2026-11-01.md`); `african_ultrachat` is UltraChat machine-translated into 11 languages, `alpaca_*` (twi, swh, zul, wol, som) are translated Alpaca sets, `amharic_*` and

kiya_sft (om, yo, sw, am rows) are machine-translated sets, and the translation system is not stated in the dataset cards or in the intake records; the packs carry no source field, so the tokens of each source drawn in stage B are not recorded.

- **Unspecified-licence sources** in the main output: KwekuAinoo/eng_twi 551,358; sentence-transformers/parallel-sentences-opus-100 210,479; UBC-NLP/AfroLingu-MT 207,796; cibfaye/wolof-english-bible 11,415; webfaq_v2; bambara_lm_qa (381k, 83% of Bambara instruction rows). Religious text dominates many low-resource languages (28 of the 81 RL languages are religious-only in the post-editor selection); the Efik NT sheet (7,936 verse pairs) has unconfirmed provenance and licence.
- **IbomMT** is CC-BY-NC-4.0: a disclosed non-commercial exception in the SFT (once) and a human-reference source of the RL reference domain (efi, ibb).
- **Quarantined and not used:** yigagilbert/synthetic-parallel-external, fiifinketia/pristine-twi-english, muri-it, the Aya parallel pairs.
- **Gemma base model:** google/gemma-3-270m-it; the Gemma terms of use apply to the derived weights; pretraining text is unknown, so exposure of benchmarks through pretraining cannot be excluded.
- **FLORES+ dev train half** (440 ids; 23,706 of 23,760 rows found in the audit) is in the SFT on purpose; the held-out half (557 ids) is protected.

N.2 RL prompts and reward-side data

- General prompts: FineWeb-Edu (ODC-By) and Wikipedia (CC BY-SA). News prompts (30% of the draw): BBC News 2017-2025 (RealTimeData/bbc_news_alltime, **no licence declared, BBC copyright**), permutans/c4-bbc-news (ODC-By label inherited from C4, BBC copyright applies), Wikinews (CC BY 2.5), Global Voices (CC BY 3.0 per the sister card), Wikivoyage (CC BY-SA 4.0; 188 prompts drawn in stage A) and SetFit/bbc-news (37 pool rows, none drawn, licence not recorded). The prompts are RL **input** text only; the source is never a training target.
- Seven new-language prompt files of stage A (tll mfe guw lub crs zne lue) had not been filtered; stage A drew 6 flagged prompts of 28,120 (0.021%), 6 of 1,345 for those languages, source side only. B1 was restarted from its step-20 checkpoint with the filtered pool.
- Reference domain (stage A from step 41, B1, mixed, science RL): human pairs from AfriScience-MT **train** (Apache-2.0; amh hau lug nso yor zul) and IbomMT train (CC-BY-NC-4.0; efi, ibb), anchors for 7 languages only. Test papers are in no pool. 0.9% of the AfriScience train rows have their African-language sentence in the A0 pair data (Appendix N).
- **NLLB-3.3B pseudo-references (CC-BY-NC) are a reward signal** for kea kbp nus bam kab (weight 0.25, max 0.35), reward only; no NLLB text is a training target in RL.
- Reward models: masakhane/africomet-qe-stl-1.1, McGill-NLP/ssa-comet-mtl; held-out monitor masakhane/africomet-stl-1.1 (apache-2.0 verified); GlotLID language guard. Licences as declared on the Hub (read 2026-11-05): McGill-NLP/ssa-comet-mtl and ssa-comet-qe apache-2.0; masakhane/africomet-qe-stl-1.1 apache-2.0; cis-lmu/glotlid license: other with license_name: apache-2.0-plus-notices (card: Apache License 2.0 plus notices).
- Science stages: science RL prompts clean_sci_v2 (PMC, arXiv CC BY/CC0, AfriScience-MT train English); **LibreTexts was excluded** from the science RL (licence conflict CC BY against CC BY-NC-SA, 10,257 prompts). **LibreTexts is not absent from the long-document RL pool**: 3,428 common-pile/libretexts rows (17.8% of science_b2048). The rows were removed before the stage started; the long-document RL did not use them. The E-14 run (science RL started from the mixed best) used the same pool and reward as the executed science RL. Science SFT: AfriScience-MT train (Apache-2.0; 28,750 windows) plus replay cut from the general SFT shards (so the NLLB, Aya and Tumbuka shares apply to the replay); 54 further catalogue candidates were scanned and none used.

N.3 TranslatePsy: used only in a research control

qvac/TranslatePsy-AfriSLM-Synthetic-Mix (Gritta et al.; CC-BY-NC-4.0; targets are facebook/nllb-200-3.3B translations of MADLAD-400 sentences, filtered by AfriCOMET, SSA-COMET and MetricX) was authorised by the authors on 2026-11-04 **for the matched-compute**

continued-SFT control (E-03) only. Its English-to-X rows (19 languages: afr amh hau ibo kin lin lug mlg nya orm sna som sot swh tsn wol xho yor zul) enter that control run, scanned against the protected index (any hit on either side removed; 0.67% of the 950,000-row sample); **no TranslatePsy row is in the SFT, any RL stage, the science or long-document stages, or any released weight.** The control checkpoints are research artefacts (CC-BY-NC-4.0 and NLLB terms), are not pushed to any public repository and are not used in production. The comparison RL against control must state two things: the TranslatePsy quality filters are the reward family of the RL, so neural metrics favour the control while chrF and chrF++ do not; and TranslatePsy targets are NLLB-3.3B-like, so a good control result partly means "distilling NLLB". Status: run (preparation `reports/sft_control_prep_2026-11-04.md`; execution record and result `reports/e03_sft_control_2026-11-05.md`): 277,900 pairs per language for the 19 languages (5.28 M pairs, 96,138 packs, each seen once) after removing every row with a hit against the protected index; the control checkpoint is a research artefact under CC-BY-NC-4.0 and the NLLB terms, kept on the authors' storage, never released and never used in a released model. The comparison states that the TranslatePsy filters are the reward family of the RL, so AfriCOMET and SSA-COMET-QE favour the control, and that its targets are NLLB-3.3B outputs (Section 5.4).

N.4 Analysis decisions and data-audit facts

- The 385M SFT text contains no benchmark sentence of 3 or more words; one- and two-word strings, numerals and names occur (NTREX 0.25%, MAFAND 0.47%, SmolSent 0.12%, GlotStoryBook 0.44% of rows); phrase-level exposure is in Appendix N.
- The RL dev set is a FLORES+ dev subset (720 sentences, 18 languages). The weak-language choice used devtest-derived per-language tables (selection on a test set).
- Benchmarks were evaluated many times; checkpoint choice used only the 720-sentence dev set. The science alpha rule was written before the headline metric was fixed; the headline then changed from SSA-COMET-MTL (reference-based) to SSA-COMET-QE after the calibration. The post-hoc RL-versus-SFT decomposition ran after the results were known.
- The earlier 270M pipeline (not the clean 385M rerun) trained on text that held the English side of 95% of FLORES+ devtest; the old 270M models; their numbers appear only on their clean rows (48 sentences per language on FLORES+) and are descriptive; the earlier 270M work's FLORES+ results are not reused.
- FLORES+ and FLORES-200 references differ in places; the NLLB comparison uses published FLORES-200 numbers.

N.5 Process disclosures

- **Gates overridden by the authors.** The RL preflight gate failed (gradient norm 3.40 > 3.0, greedy loop 0.109 > 0.02, greedy finished 0.906 < 0.97) and was overridden; the stage-A gate failed (max KL 0.169 > 0.15, three KL-guard cuts, greedy loop 0.016 > 0.01) and B1 was started on the authors' instruction.
- **Mid-run changes.** Stage A was stopped at step 40 and resumed with the four features; B1 was restarted at its step-20 checkpoint (filtered prompt pool, reference domain, oversample 1.0, KL guard 0.25) and again at step 40 (MBR guide, dev early stop); the mixed stage was restarted at its step-20 checkpoint with the patch-3 monitor, and the authors loosened the hacking alarm (chrF++ 0.3 $\rightarrow$ 0.75; AfriCOMET 0.005 $\rightarrow$ 0.03). Stage A's KL guard cut the learning rate three times and it never recovered.
- **Start checkpoints.** Mixed started from B1 best (step 50), not B1 final; B1 final exists on the Hub and was evaluated after the fact (E-20): it equals B1 best within noise (+0.01 chrF++ [-0.02, +0.04] over 93 cells). The weak-group run never produced a `best/` checkpoint; its final (step 85) was evaluated. The `masakhane/africomet-stl-1.1` monitor influenced early stopping in the mixed stage and later (never a reward).
- **Incident.** The mixed stage was auto-started once by a process outside the launch agent, stopped within a minute without a step, and relaunched from B1 best. A first science-SFT pipeline pass wrote an empty alpha and launched the science RL with an invalid model path, which stopped itself at start. The GPU idled for about 2 h 10 min between stage A and B1. The E-14 run (science RL from the mixed best) had one CUDA out-of-memory in the COMET reward worker at about step 38; the launcher lowered the

micro-batch from 12,288 to 6,144 tokens and resumed from the step-20 checkpoint, so steps 21 to 38 were replayed (same prompt schedule, vLLM sampling not bit-reproducible); `metrics.jsonl` holds two records for those steps and the later one is used.

- **Science RL length.** The science-RL yaml names 150 steps a hard cap, but the trainer's extension rule raised the cap to 190 steps at step 120 while dev metrics were still improving; the run then stopped on a dev plateau at step 150 and, after a 10-step wind-down, executed 160 steps (`summary.json`). The long-document RL has the extension rule off.
- **Checkpoint storage.** Stage checkpoints were written to RAM-backed `/dev/shm`; weights are in anonymised repositories, optimiser states are not.
- **Evaluation-only tool under a non-commercial licence:** the NLLB-218 language-ID model (cc-by-nc-4.0) was used for measurement only.
- **Compute.** A100 80 GB (vast.ai) for SFT stage B (44.8 h), RL (8.66 GPU-h for A + B1 + mixed), evaluation and the science stage; smaller boxes for data preparation and the post-editor. GPU-hours by component, estimated from logged durations, are in Appendix (compute table): the sized components sum to about 94 A100-hours; A0, SFT-1 and several evaluation passes are not sized; no cost figure is reported.

N.6 Intended use and risks

Outputs are machine translations of unverified quality for languages without any human evaluation, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable (chrF++ 2 to 14 after RL), RL increases English output there, and the model should be used sentence by sentence, not on whole documents. Religious text dominates the data of many low-resource languages, which may shift register and vocabulary toward it. The model must not be presented as a replacement for NLLB-class systems: it is 3.26 chrF++ below NLLB-3.3B on 43 shared languages. The authors' use case is production translation; the licences of NLLB and TranslatePsy prevent using those systems there, which is why we state exactly where NLLB-derived text and signals entered this model.

O Reproducibility statement

- **Code and configs.** `gemma_mt/` (trainer `train_sft.py`, RL `gemma_mt/rl/`, evaluation scripts); SFT configs `reports/data/e23_hygiene_2026-11-04/sft_configs/`; RL configs `.../rl_configs/` (note: the archived stage-A `config.json` is the resumed configuration); reports and per-language tables under `reports/` and `reports/data/` as keyed in `CLAIMS_LEDGER.md`; analysis scripts for every new section in `reports/data/{nllb_comparison_rl_models,family_script_resource_analysis,degeneration_audit_independent,stage_A_feature_split,contamination_followups_385m,e23_hygiene}_2026-11-04/`.
- **Seeds and hyper-parameters.** Seed 1234 throughout; SFT: Muon, peak 5e-5 (SFT-1, shard 1) and 3e-5 (shards 2-6) with cosine floors 50% (SFT-1), 80% (shards 1-5) and 10% (shard 6), 131,072 tokens per step (Section 3); RL stages A to mixed: Muon, 2e-6, group 16, KL 0.05 (reference = each stage's start model), guard 0.15/0.25, temperature 1.0; **science RL: Muon, 1.0e-6, KL 0.1, group 16, guide 0.25, 160 steps executed (yaml cap 150, extended to 190, stopped by dev plateau); long-document RL: Muon, 1.0e-6, KL 0.1, group 8, no guide, 80 steps $\times$ 32 prompts** (Table 17); science SFT AdamW 3e-6, alpha 0.5. The science-RL yaml is in `reports/data/rl_configs_as_executed_2026-11-04/sci_rl_F_final.yaml`.
- **Hardware and software.** One A100 80 GB; vLLM 0.32.0 bf16 greedy, 768 new tokens, for every headline number. Outputs are not bit-identical across engines (9.6% identical between vLLM bf16 and HF bf16 on 240 sentences); a reader reproducing with another engine should expect small, unsystematic differences (240 sentences, HF bf16 minus vLLM bf16: +0.54 [−0.67, +1.79] chrF++ for the SFT and −0.23 [−0.73, +0.23] for the RL model; the SFT in fp32 against bf16 on the full set: −0.04 [−0.13, +0.07] over 93 cells, E-09b, which supersedes the earlier +0.24 [−0.07, +0.55] on 4,000 sentences and +1.81 on 240 sentences). sacreBLEU 2.6.0 and the chrF++/spBLEU settings are given in Section 4; the full signature strings are not recorded; exact commands per table are in the archived scripts named in

the ledger.

- **Weights and branches.** RL repository https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl: branch main (mixed best, the final general RL model), latest, stage-a-b64 (stage A step 200), stage-a-final (step 400), stage-b1-b256 (B1 best, step 50), stage-b1-final (step 135), stage-g-weak-latest, science-sft, science-sft-alpha1, science-rl, science-rl-latest, science-rl-final, long-rl, long-rl-latest (long-document RL checkpoints) and tag rl-step-50. SFT final: https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft (branch main, step 32,853). Experiment weights: https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments, branches e04-seed2-step100, e04-seed3-step100, e04-a_base-step100, e04-c_noref-step100, e04-d_noguide-step100, e14-science-rl-only-final. The control checkpoint of E-03 is not published (CC-BY-NC data). No licence field or model card is declared on these repositories **[OWNER: confirm release licence]**. Run configs, logs, per-sentence outputs and scripts of E-02, E-04, E-11 and E-14 are in the research archive and the reports directory named in the ledger.
- **Data.** The decontaminated SFT and RL prompt data and the protected index (protected-2026-10-28.3) are in anonymised storage named in the reports and released on acceptance; the NC and unspecified-licence parts cannot be redistributed as they are (Appendix N); TranslatePsy rows are not released (Appendix N). The state, prompts, eval and reports repositories were private at the last check (Hub API 2026-11-04). [OWNER: which parts of the data are released.]
- **Long-document RL incident.** The stage took 7.7 h of wall-clock time (10:03 to 17:44 UTC), of which about 4.5 h were a crash loop: the trainer stopped (ENOSPC, root disk full) at the first step-25 evaluation and Hub-staging save and was restarted 18 times by the supervisor, each time resuming from last/ at step 20 and repeating steps 21 to 25; the staging directories were real directories on the root disk instead of symlinks to /dev/shm. Fixed at 15:35 UTC; the model path of the 80 steps is otherwise the planned one (one seed). About 3.2 h were productive. The repeated step-20-to-25 segments are visible as repeated step-25 evaluations in the metrics log.
- **Not reproducible from what is stored:** stage-A steps 1-40 hyper-parameters except through the stage report; B1 final's evaluation; per-sentence NLLB-3.3B outputs on NTREX and MAFAND (only corpus-level aggregates survive); the training seed spread; the SFT with another seed (stage B alone is 44.8 A100-hours).

P Prompt format and data statement

Prompt format. Every translation request is the fixed prompt Translate the text to {language}: {english} (two spaces after the colon) inside the Gemma 3 chat template; the model answers <tag> {translation}<end_of_turn> with the single-token tag of the target language; the loss covers the answer only (Section 3). **RL prompts** are English windows of consecutive sentences from FineWeb-Edu and Wikipedia (70%) and from news (30%), filtered against the protected index; 81 languages with higher draw weights for the 29 weakest (Section 3).

Data statement. Training data are English-to-African-language pairs and instruction data in 125 languages from the sources listed in Ethical considerations and Appendix N; many low-resource languages are dominated by religious text; stage B includes NLLB-mined, Aya-aligned and machine-translated instruction data; evaluation data are human references of FLORES+ devtest, NTREX-128, MAFAND-MT, SmolSent, SALT, GlotStoryBook, IbomMT, FLORES-MOZ and AfriScience-MT. No personal data was collected by us; no human subjects were involved; no human judgement of system outputs was collected.

Q Responsible NLP checklist: items filled from the paper

- **Scientific artefacts used and licences.** Base model: Gemma-3-270m-it (Gemma terms of use apply to the derived weights). Reward and monitor models, as declared on the Hub (read 2026-11-05): McGill-NLP/ssa-comet-mtl and ssa-comet-qe, masakhane/africomet-qe-stl-1.1 and africomet-stl-1.1: apache-2.0; GlotLID (cis-lmu/glotlid): license: other,

`license_name: apache-2.0-plus-notices`. Training and reward-side data and their licences are listed in Ethical considerations and Appendix N: NLLB-mined and NLLB-derived data (CC-BY-NC terms), IbomMT (CC-BY-NC-4.0), AfriScience-MT train (Apache-2.0), FineWeb-Edu (ODC-By), Wikipedia (CC BY-SA), several sources with unspecified licences; TranslatePsy (CC-BY-NC-4.0, NLLB-3.3B targets) only in the research control. Evaluation uses the public benchmarks named in Section 4; the NLLB-218 language-ID model (CC-BY-NC) was used for measurement only. Release licence of the weights: the Hub repositories declare none **[OWNER: confirm release licence]**.

- **Parameters and compute.** 385,144,448 parameters; one A100 80 GB at a time. Table 24 gives A100-hours per component, *estimated from logged durations* (training seconds from the run summaries, or the logged start and end times of the job); no cost figure is reported. The sized components add up to about 94 A100-hours, a lower bound. **Not sized** (no duration in any report): the A0 stage (5,625 steps) and SFT-1 (1,500 steps); E-01 (guard grid and guarded full evaluations); E-09 and E-09b (the fp32 generation ran 17:46 to 18:38 UTC beside another job at a 0.15 GPU share); E-13 and E-20; the SSA-COMET scoring of E-05; the full evaluation passes of every stage (122,585 rows each) and the AfriScience and document evaluations; data preparation and scoring boxes; aborted runs and the compute of the earlier 270M models.
- **Number of runs and seeds.** One training run per model and stage (seed 1234); two extra seeds (20261105, 20261106) of the stage-A recipe at 100 steps; three ablation arms, one run each; the control and the E-14 run are one run each. No multiple-run statistics beyond that pair; test-sentence bootstrap intervals elsewhere.
- **Hyper-parameters and selection.** Appendix C (all SFT and RL settings, per-shard schedules, reward and features). Checkpoints were selected on a 720-sentence FLORES+ dev subset; the decode-time guard setting by a rule fixed in advance on the SFT dev grid.
- **Evaluation.** Metrics defined in Section 4; sacreBLEU 2.6.0 (from the install log of the evaluation machine; the full signature strings were not recorded), chrF++ = `CHRF(word_order=2)` with the defaults, spBLEU = BLEU with `tokenize=flores200`, NFC-normalised hypotheses and references; vLLM 0.32.0, bf16, greedy, 768 new tokens; intervals are 95% paired bootstrap, 1,000 resamples (language bootstrap, 10,000 resamples, for NLLB). Exact commands per table are in the archived scripts named in the ledger.
- **Human subjects and annotators.** None were involved; no human judgement of system outputs was collected; no personal data was collected by us.
- **Code, data and checkpoints.** See the links in Ethical considerations and the branch list in the reproducibility statement, subject to the licence limits above. NLLB-derived and non-commercial parts cannot be redistributed as they are, and TranslatePsy rows are not released **[OWNER: which parts of the data are released]**.

Table 24: A100-hours by component, estimated from logged durations (lower bound; unsized components are listed above).

component	A100-h	basis
SFT stage B (31,353 steps)	44.8	5.14 s per step
RL stage A, B1, mixed (400 + 135 + 60 steps)	8.66	15,920 + 10,479 + 4,786 s
weak-group RL (85 steps)	2.1	7,642 s
science SFT (283 steps)	0.3	17 min
science RL (160 steps)	4.8	17,153 s
long-document RL (80 steps)	7.7	wall-clock 10:03–17:44, of which about 4.5 h crash loop
E-03 control (6,000 steps and two probes)	10.5	GPU session 9.75 h, evaluation 0.55 h, SSA-COMET pass 0.2 h
E-02 beam-4 and MBR-16	2.6	logged GPU time
E-04 two seed runs and their evaluations	3.2	wall-clock 08:51–12:05
E-11 three ablation arms and evaluations	2.9	36 + 54 + 42 min of training, 3 × about 15 min of evaluation
E-14 science RL without science SFT	6.0	training 15:18–20:17, evaluation about 1 h
sized components, total	about 94	lower bound